



Hardware components

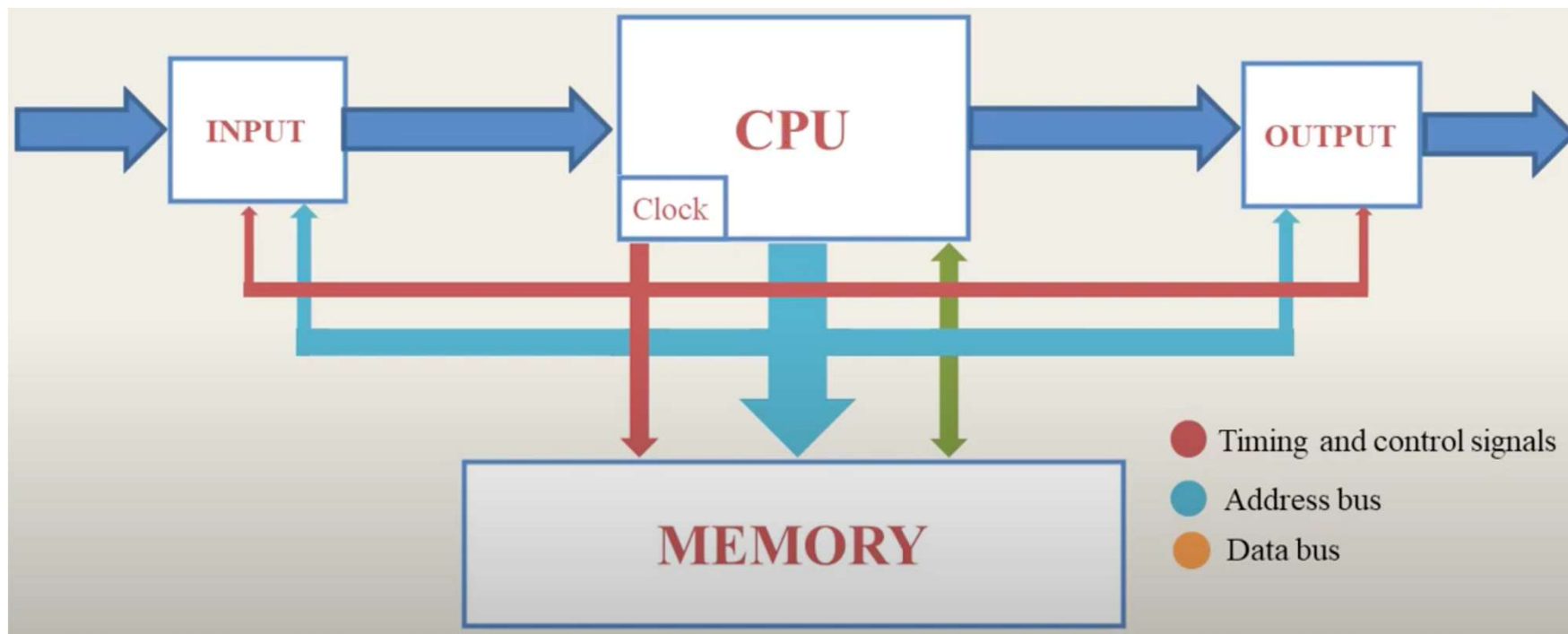
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- Central processing unit - CPU
- Memory and memory design
- Pairing with peripheral devices
- 8051 microprocessor: Architecture, memory and practical interface
- ❑ Yêu cầu ôn lại các kiến thức trong môn Kiến trúc máy tính. (Bắt buộc)

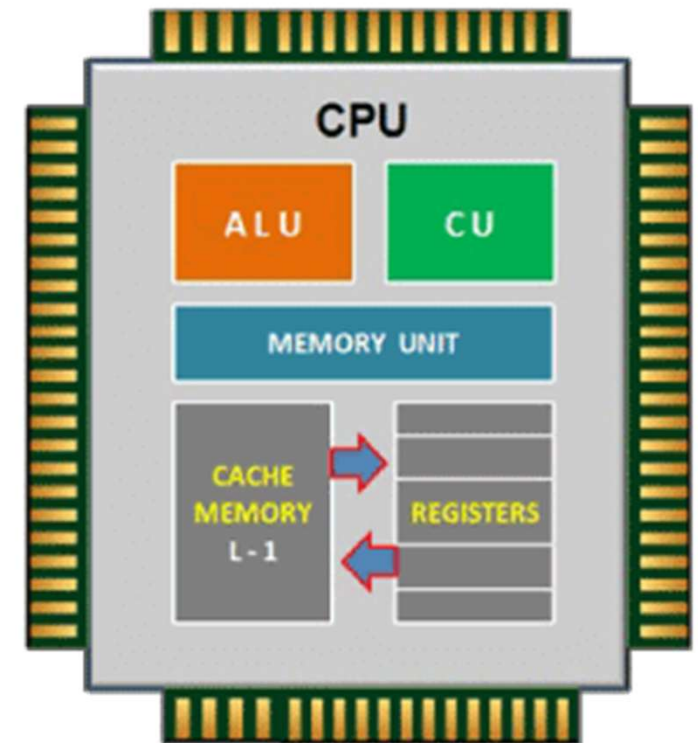


- CPU (Central Processing Unit): Executes instructions and processes data
- Memory: Program and data memory
- Peripheral: Input / output devices
- Bus: transmits signals between components

❖ Basic CPU block diagram includes:

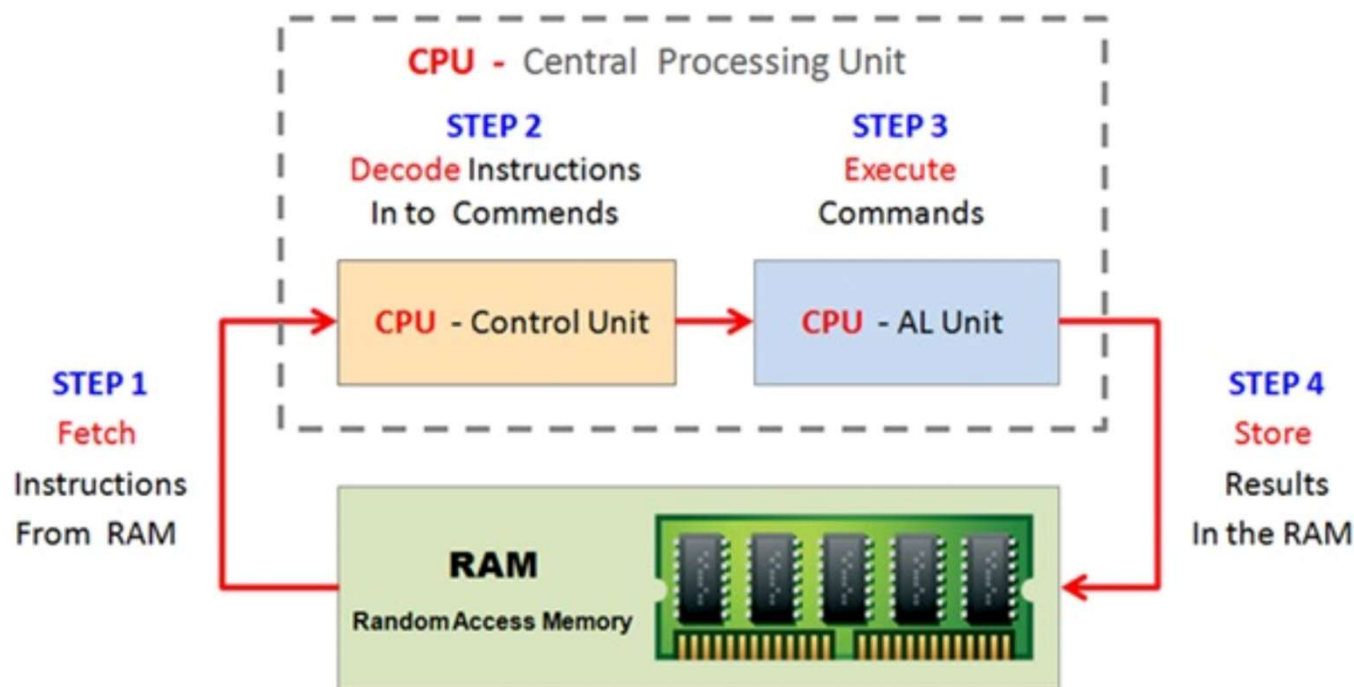
- ALU: Computational and logic unit
- CU: Control unit
- Memory Unit: Some memory is built into the CPU to support storage and fast calculations. Includes registers and cache.

Central Processing Unit



❖ Process of executing a command:

1. Load instructions from memory into CU
2. CU Decodes the command
3. The ALU executes the command
4. The results are saved in memory



❖ Instruction Set Architecture (ISA):

The instruction set architecture defines how the CPU interacts with data and memory through:

- Addressing Modes
- Data Storage Organization
- Register Sets

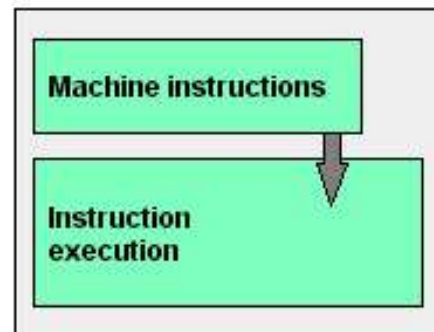
In embedded system design, the choice of instruction set architecture directly affects:

- Processing Performance
- Source Code Optimization
- Development Tools

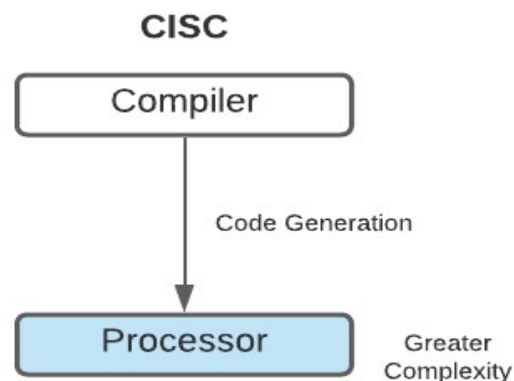
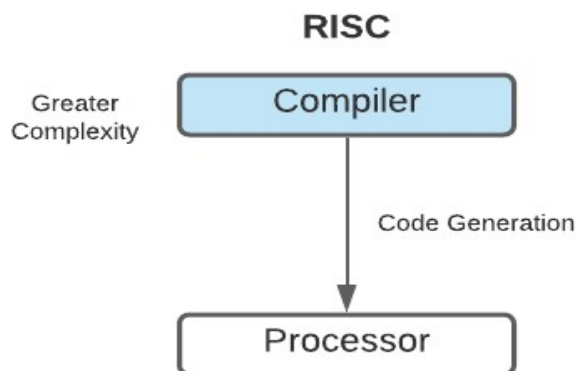
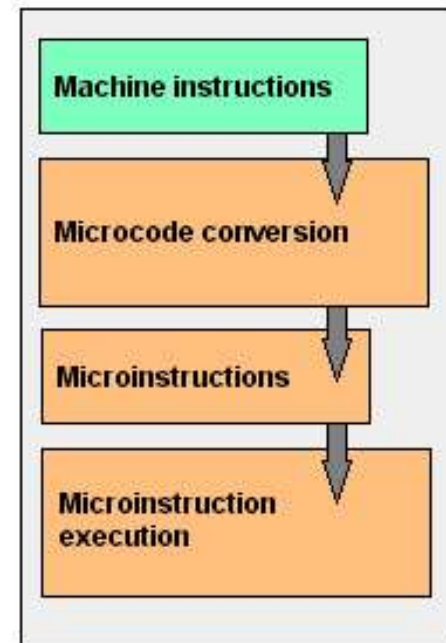
❖ ISA (Instruction Set Architecture):

- CISC (Complex Instruction Set Computing)
- RISC (Reduce Instruction Set Computing) - ARM

RISC



CISC

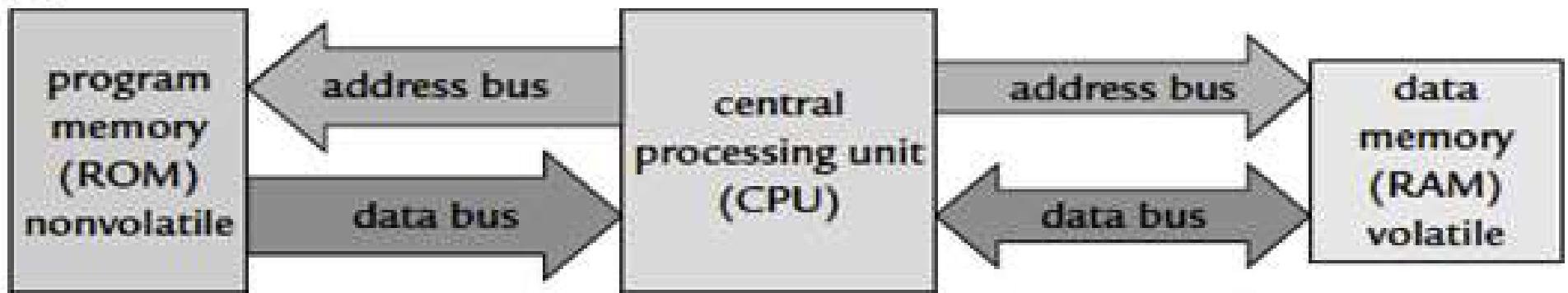


❖ Kiến trúc tập lệnh ISA (Instruction Set Architecture):

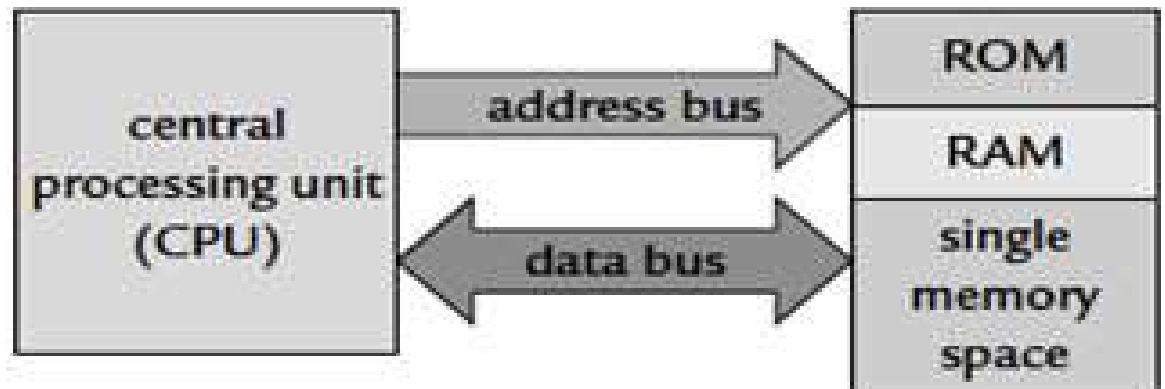
Đặc điểm	CISC (Tập lệnh phức hợp)	RISC (Tập lệnh rút gọn)
Truy cập bộ nhớ	Bất kỳ lệnh nào cũng có thể truy cập	Chỉ lệnh Load và Store
Độ dài lệnh	Thay đổi tùy theo độ phức tạp	Cố định
Thời gian thực hiện	Nhiều nhịp chu kỳ	Thường chỉ một nhịp chu kỳ
Thanh ghi	Chỉ có một tập thanh ghi	Có nhiều tập thanh ghi
Cơ chế thực thi	Thông dịch qua vi mã (microcode)	Phản cứng thực thi trực tiếp
Hỗ trợ Pipeline	Không hỗ trợ	Hỗ trợ mạnh mẽ
Kích thước mã	Nhỏ gọn, tối ưu bộ nhớ	Lớn hơn

❖ Architectures

(a) Harvard architecture



(b) von Neumann architecture

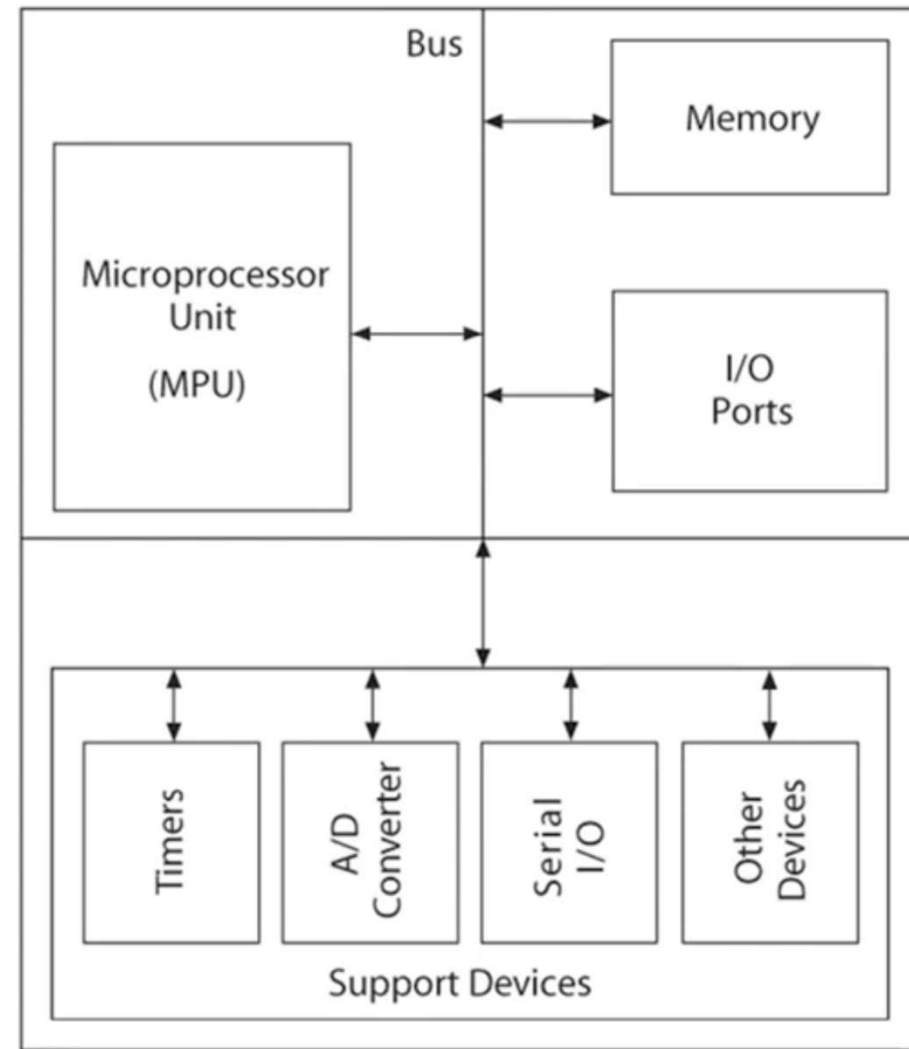


❖ Microcontroller is considered as a fully integrated circuit with components:

- Processor
- Memory
- Peripherals (GPIO, Timer, SPI..)

□ For example:

- Processor: Core i3, i5, i7...
- Microcontroller: PIC, AVR...

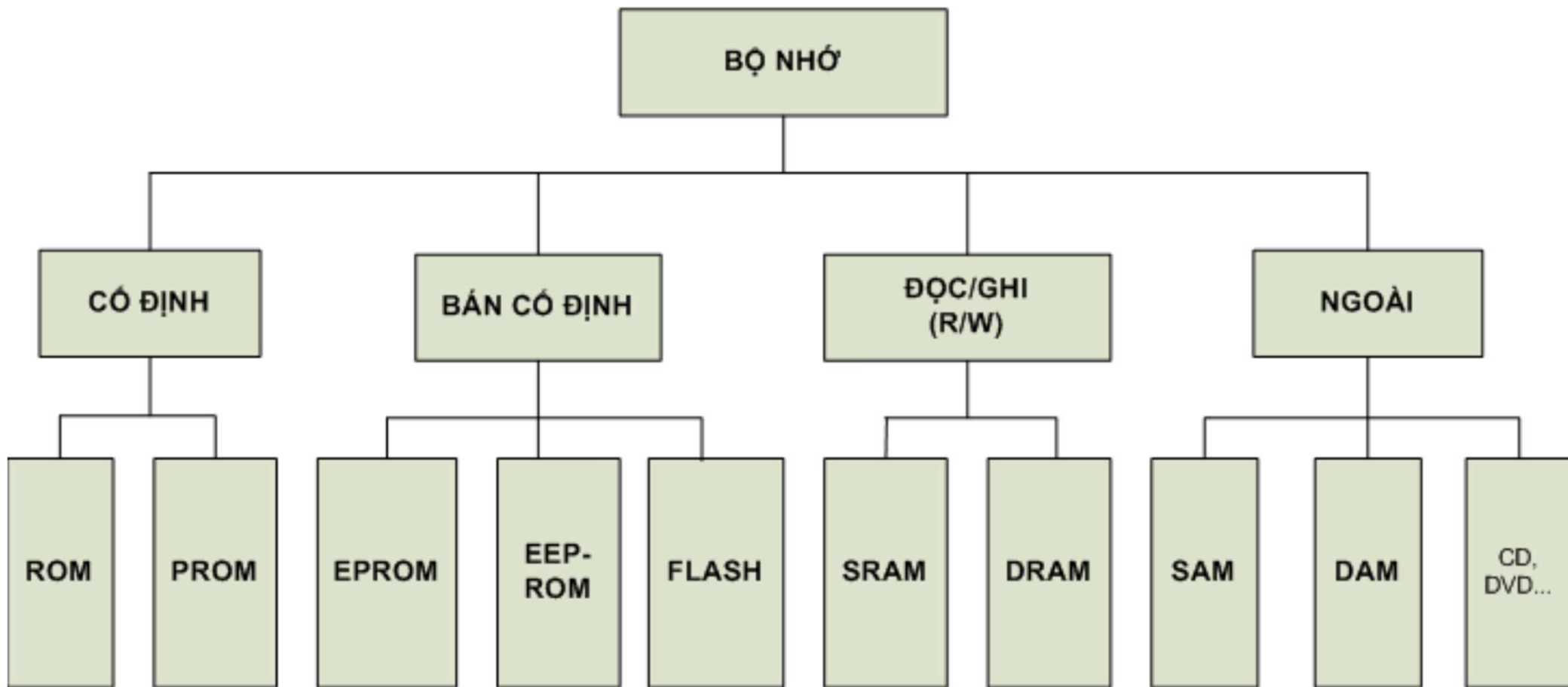


Memory and memory design

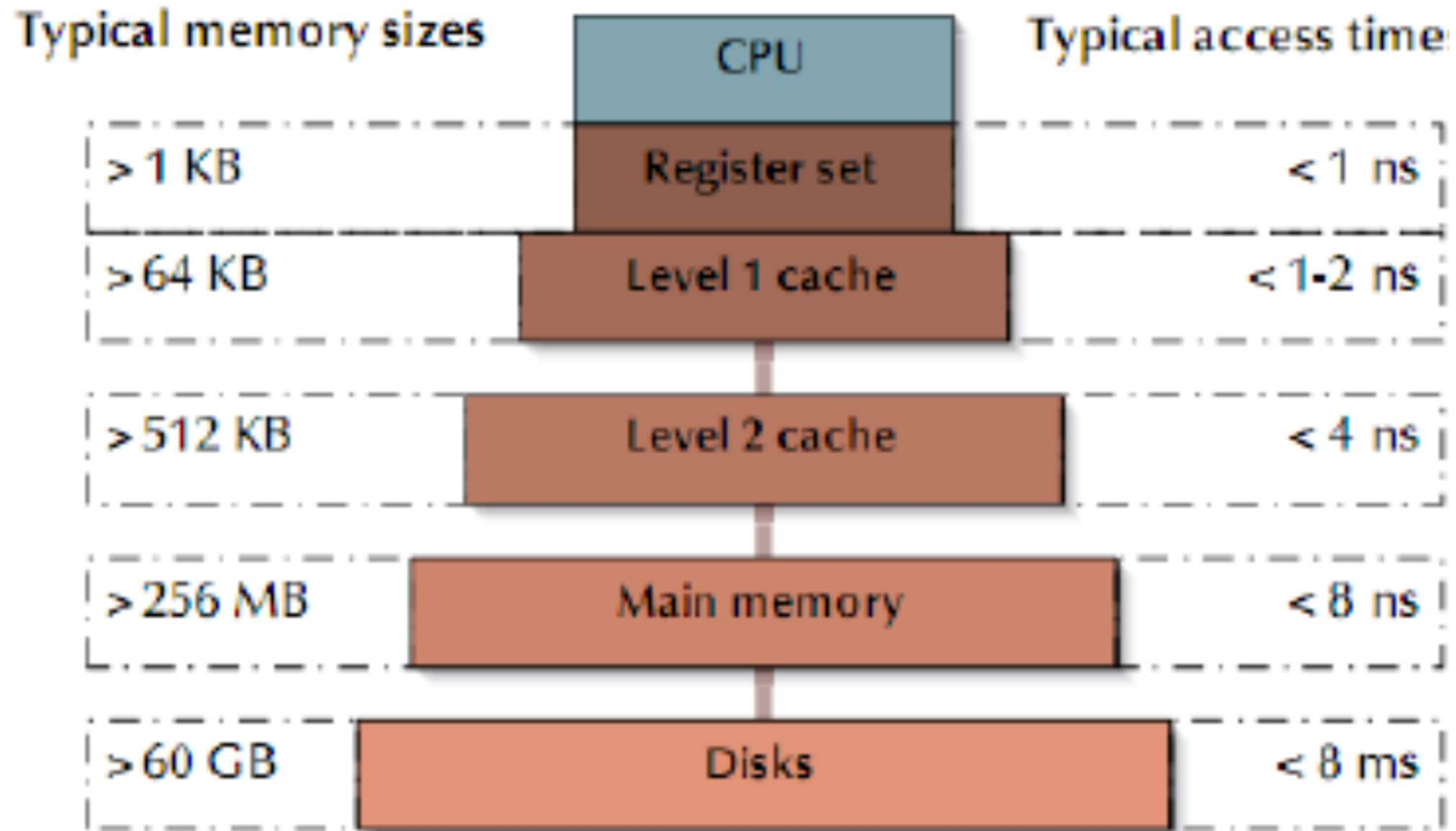
- ❖ Memory is a means of storing information including programs and data.
- ❖ Some basic information about computer memory.

Memory Type	Access Time	Cost /MB	Typical Amount Used	Typical Cost
Registers	1ns		1KB	
Cache	5-20 ns		1MB	
Main memory	60-80ns		MB	
Disk memory	10 ms		GB	

❖ Classification:



❖ Memory hierarchy model:



❖ Memory hierarchy model:

The storage system is organized in a hierarchical structure to optimize the balance between speed and cost:

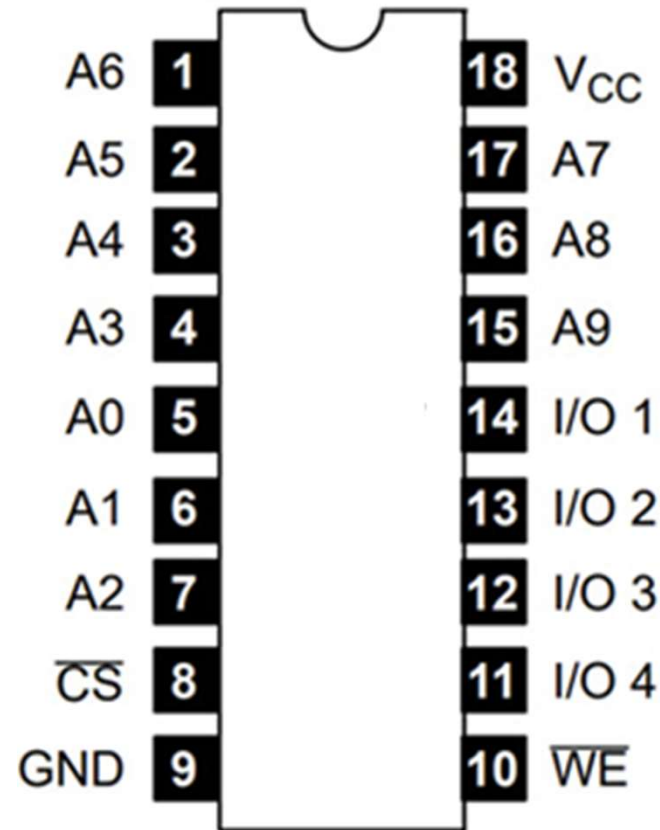
- Registers: Located within the CPU, the fastest but smallest in size.
- Cache: Located between the CPU and main memory, it contains a copy of data and is often used to speed things up.
- Main memory: Contains the program and data currently being executed.
- Secondary memory: Hard disks, USB drives, CD-ROMs, etc., store large amounts of data.

❖ Memory design

The physical memory design process involves combining discrete memory chips to create the memory capacity and word length required by the CPU.

- A memory chip is a specific microchip, arranged with basic pins.
- The pins of a typical memory chip include address bus inputs, data inputs, chip select control pins, write/read and power pins.

- For example: a 1Kx4 static RAM (1024 “memory words”, each word is 4 bits long)



❖ Memory design

■ Steps to follow:

1. Design the schematic diagram: Use address decoder circuits (such as SN74138) to determine the memory areas of the board within the CPU's address space.
2. Connect control signals: Connect the RD (Read), WR (Write) lines and the corresponding data BUS system between the CPU and the memory chips.
3. Address partitioning: Set up an address table (Memory Map) to manage which areas are allocated for ROM (containing BIOS/OS) and which are allocated for RAM (data/applications).

❖ Memory design

- Designing memory from available memory chips has the following steps:

1. Determine the number of memory chips to create the required memory capacity according to the formula:

$$M = Q/D, \text{ whereas}$$

Q is the capacity of memory.

D is the capacity of each chip

M is the number of memory chips needed.

2. Determine the number of base address wires (ie the number of low address wires connected directly to the memory chip or interconnect chip): according to the following expression:

$$2^m = D, \text{ whereas}$$

m is the wire number of the base address.

D is the capacity of each chip

3. Determine the number of address lines to create chip select according to the following expression:

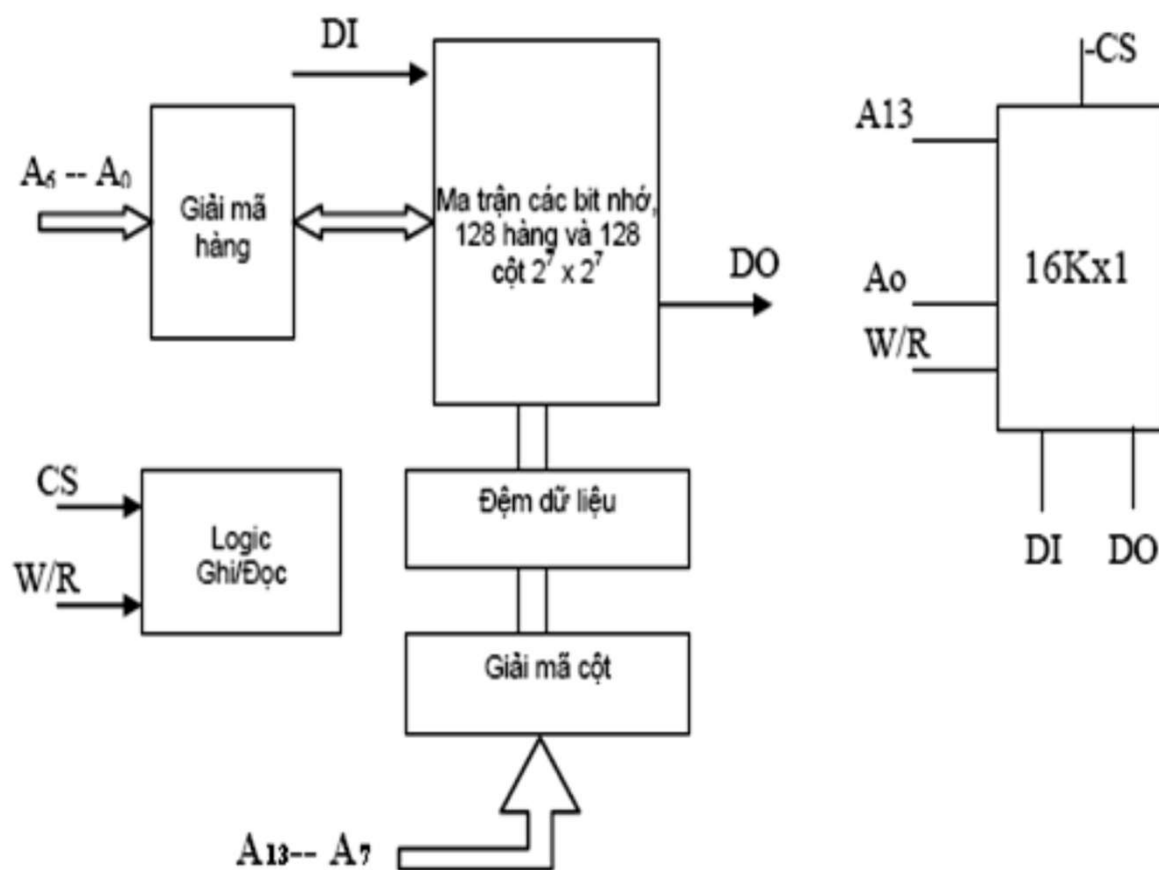
$$2^i = M, \text{ whereas}$$

D is the capacity of each chip

i is the number of wires needed to decode the signals chip select signal (Csi)

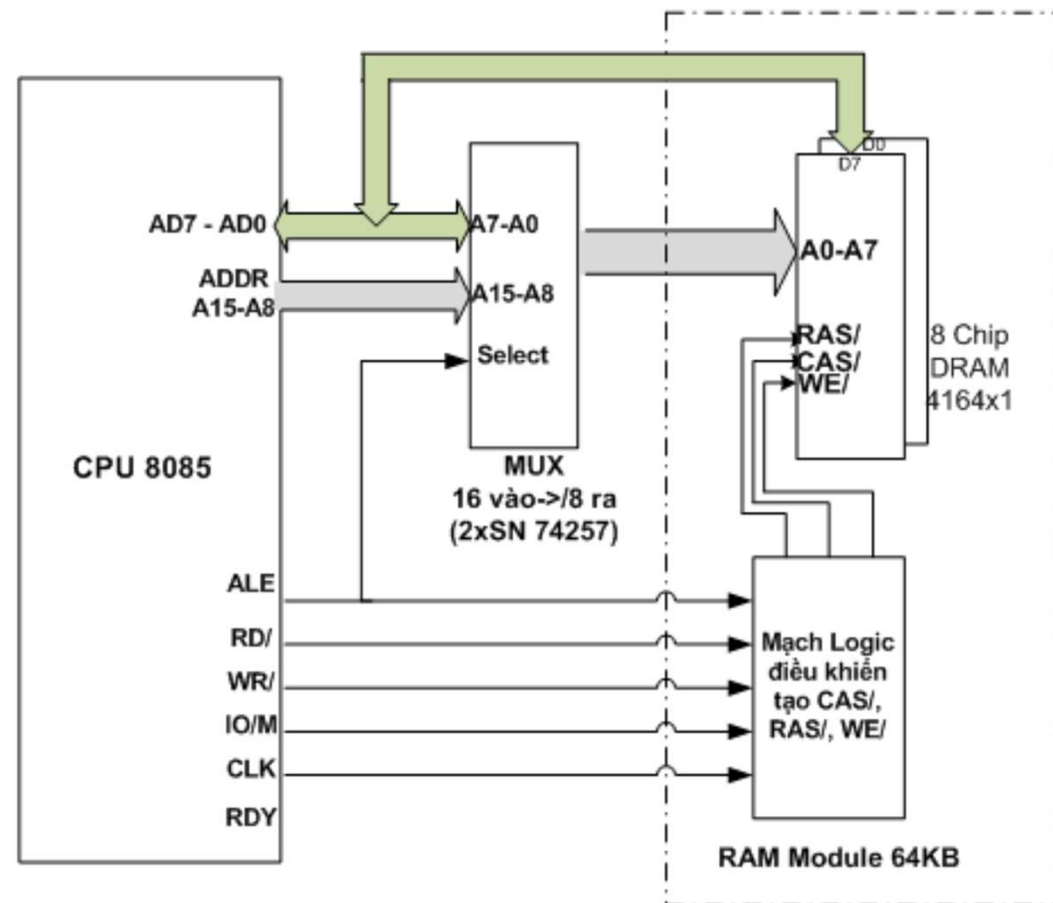
❖ Design Static RAM

Build a 16Kbyte memory based on 16Kx1bit SRAM chips. The 16Kbyte SRAM memory tape is built on the basis of 8 16K x 1bit SRAM chips, to obtain a memory cell with a length of 8 bits.



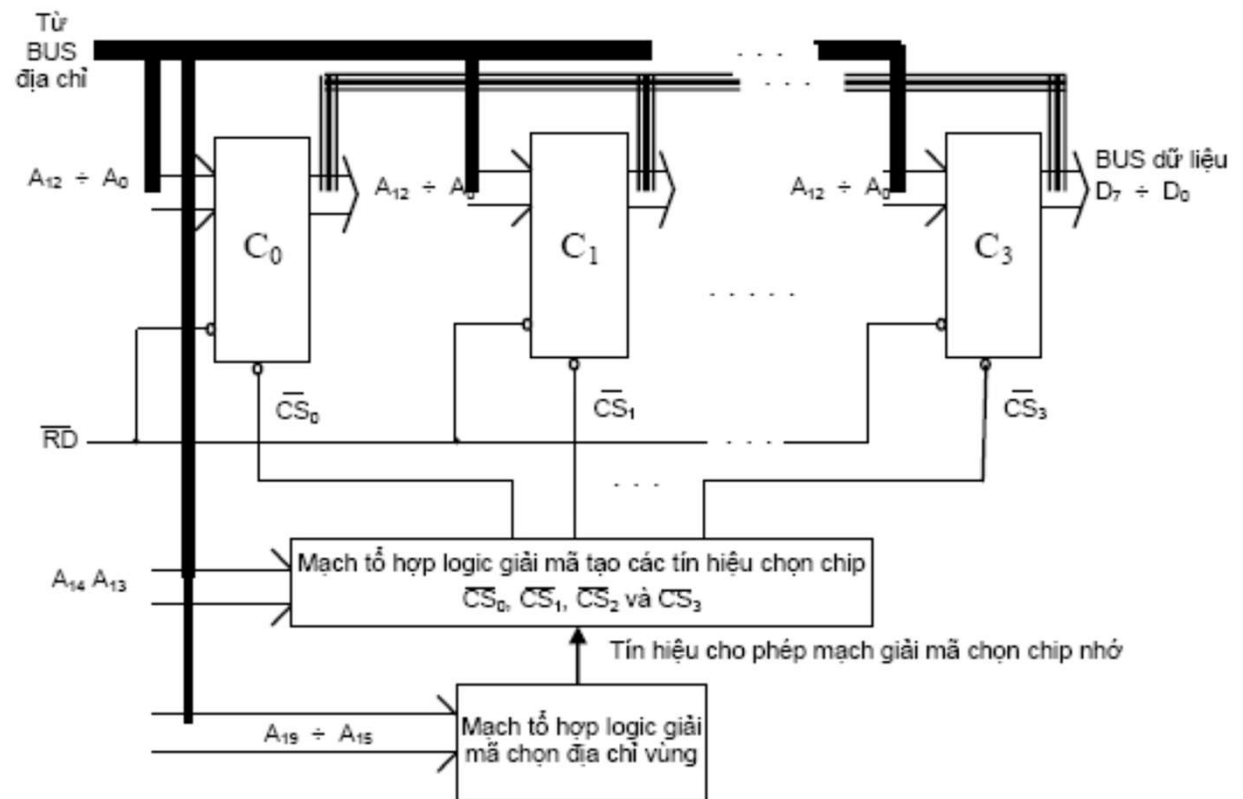
❖ Design DRAM

DRAM MK 4164 is 64K bit DRAM in 1 Chip. Assuming we will design RAM for the 8085 CPU with a maximum RAM of 64 KB, 8 Chips will be needed.

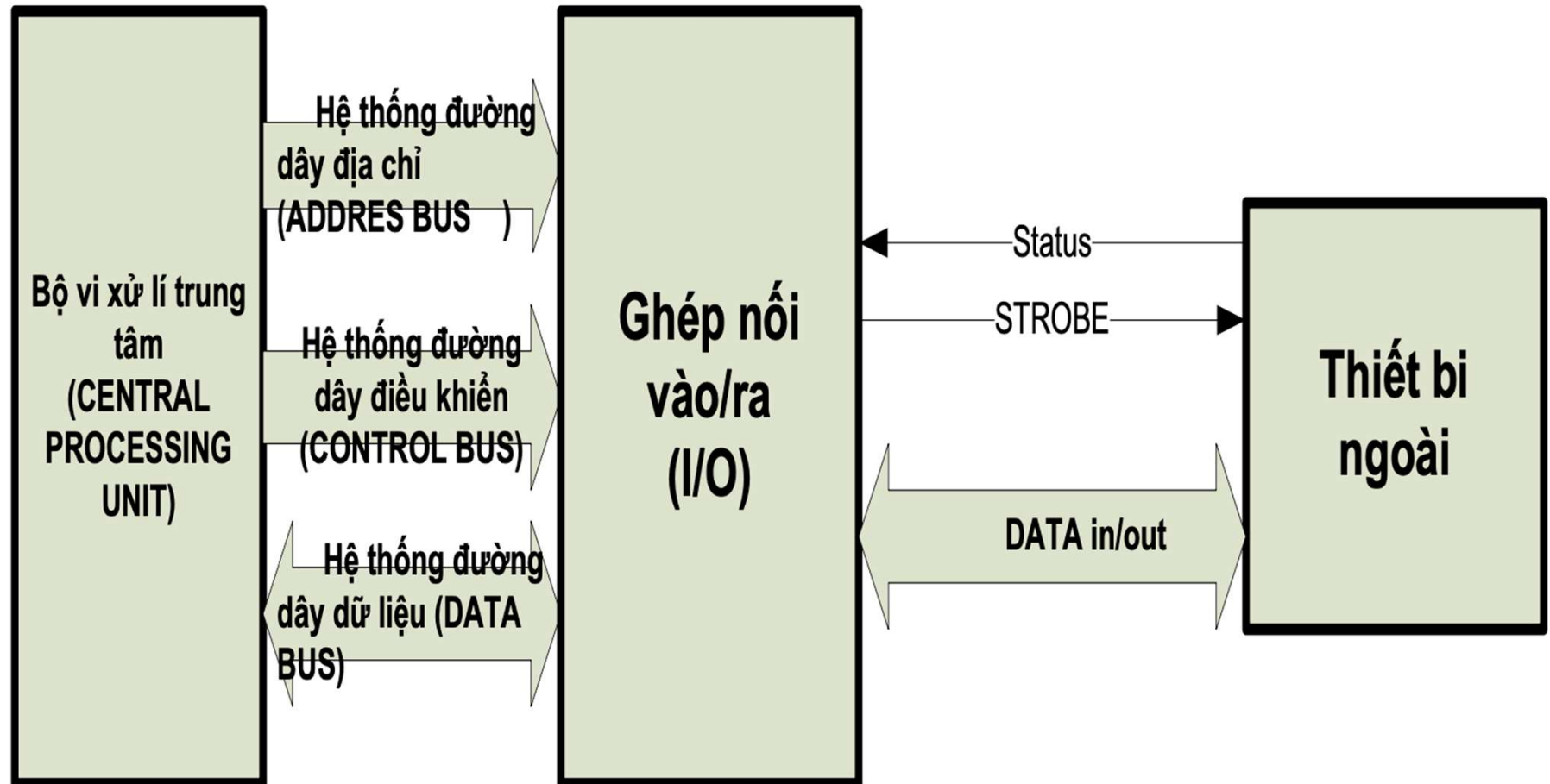


❖ ROM/EPROM

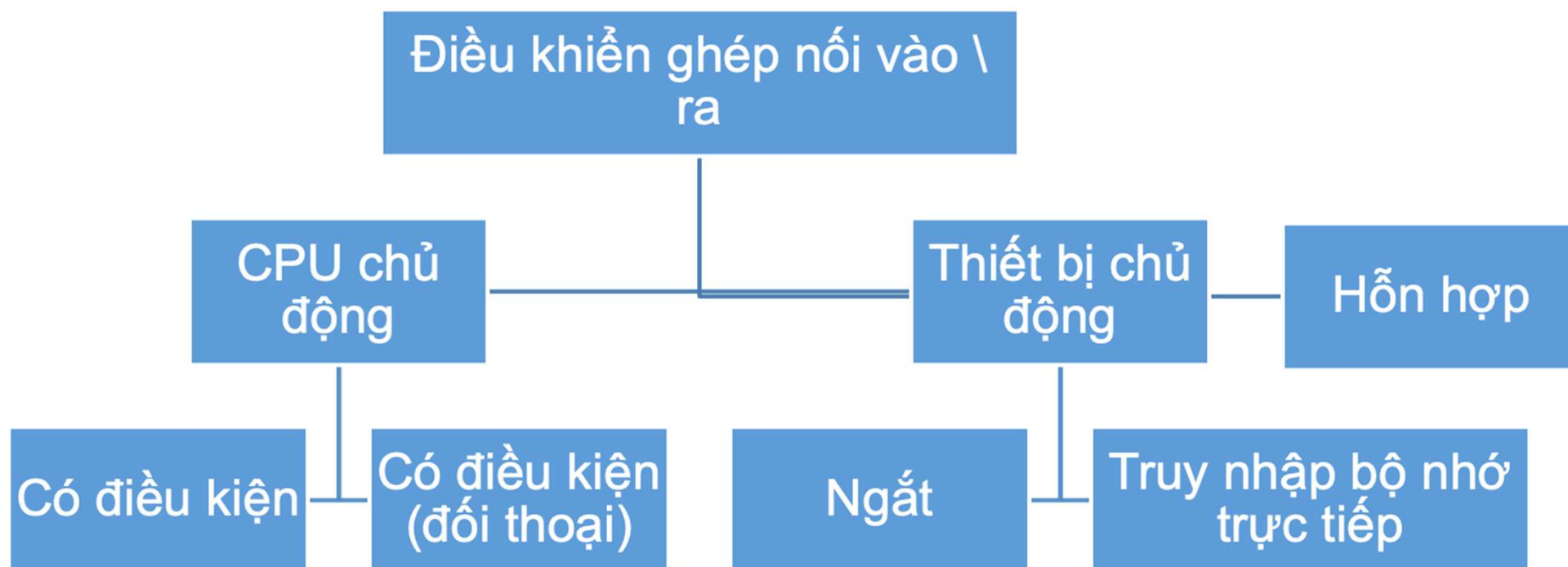
Build a ROM module with a capacity of 32KB, using Chip 2764 8K x 8 bits, the first address is 20000hex. The application program loads into this module.



❖ Model of coupling technique

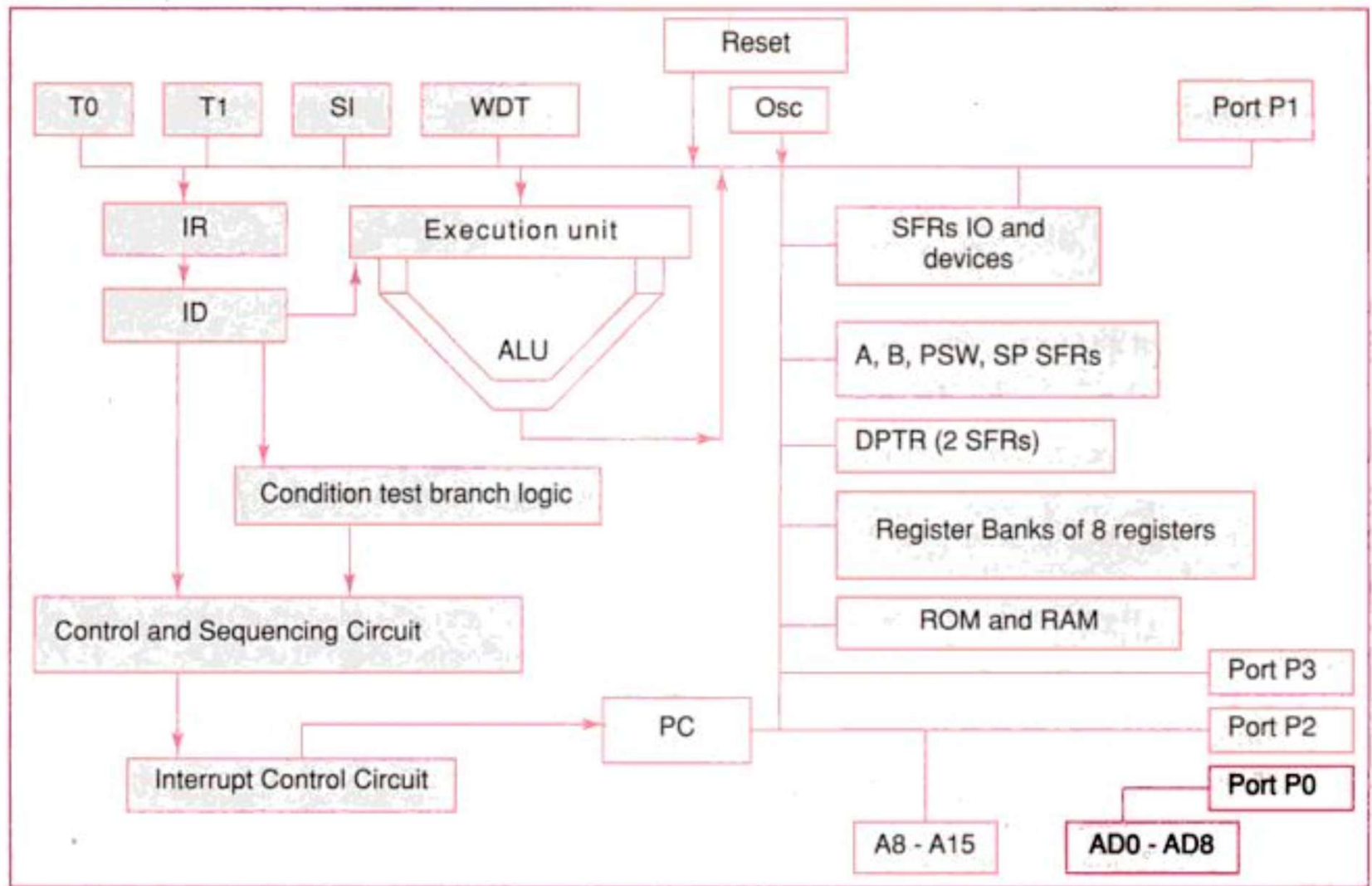


❖ Các kiểu ghép nối



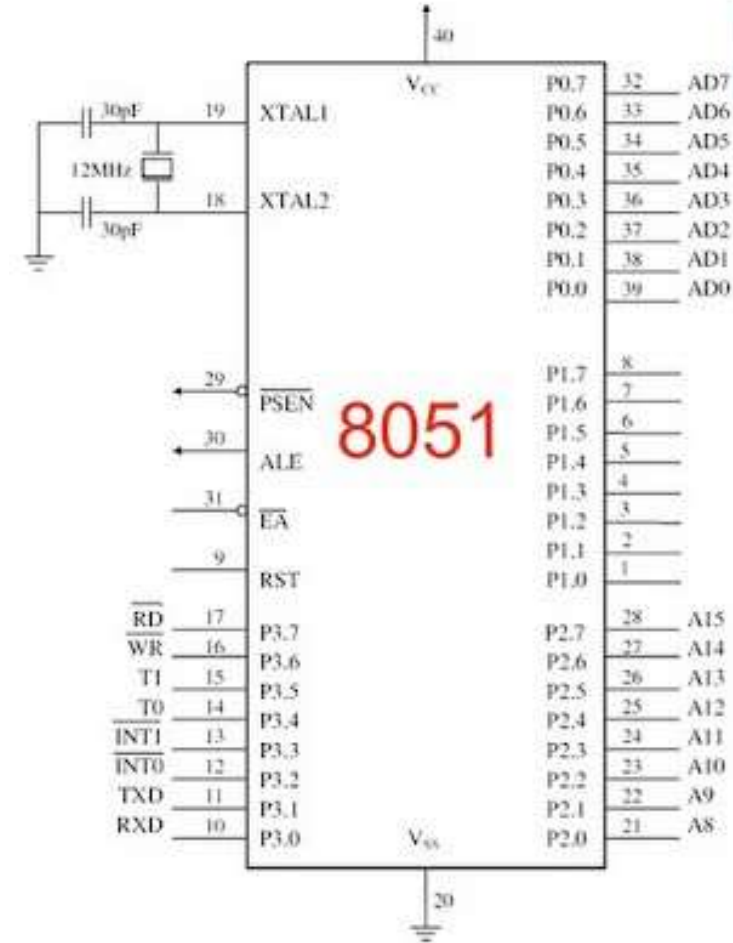
8051 microcontroller architecture

- ❖ Basic 8051 architecture: microprocessor, registers, memory (in Harvard architecture) and ports, counters/timers, I/O, and interrupt handlers.



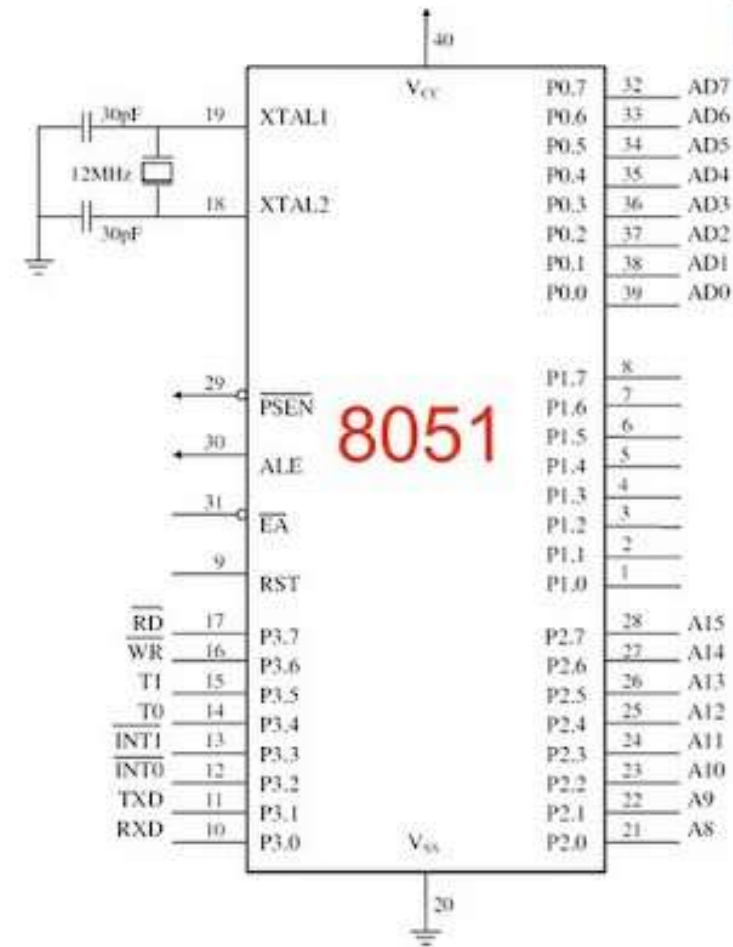
❖ Characteristics of 8051

- 12 MHz clock. Processor instruction cycle time 1 μ s.
- ALU-8bit
- Harvard Memory Architecture
- Internal 8-bit data bus and Internal 16-bit address bus
- CISC instruction set architecture



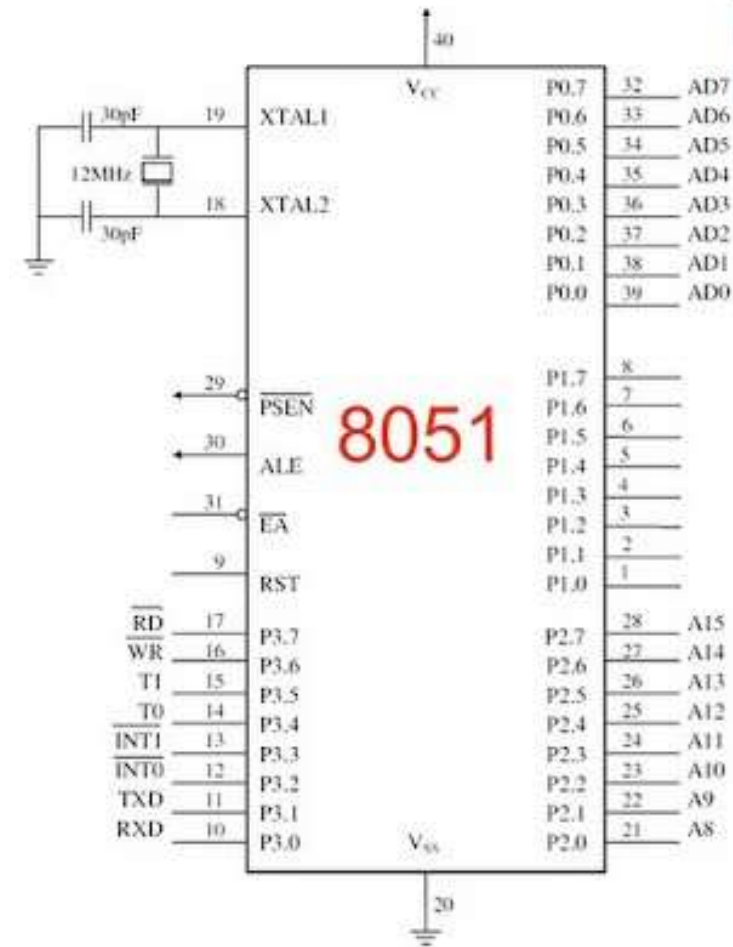
❖ Characteristics of 8051

- Special Registers (SFR)
 - PSW (status word)
 - A (cumulative)
 - Register B, SP (stack pointer)
 - Registers for I/O, timers, ports, and interrupt handlers
 - 16-bit program counter (PC) with factory default value of 0x0000.
 - 8-bit stack pointer (SP) with an initial default value of 0x07



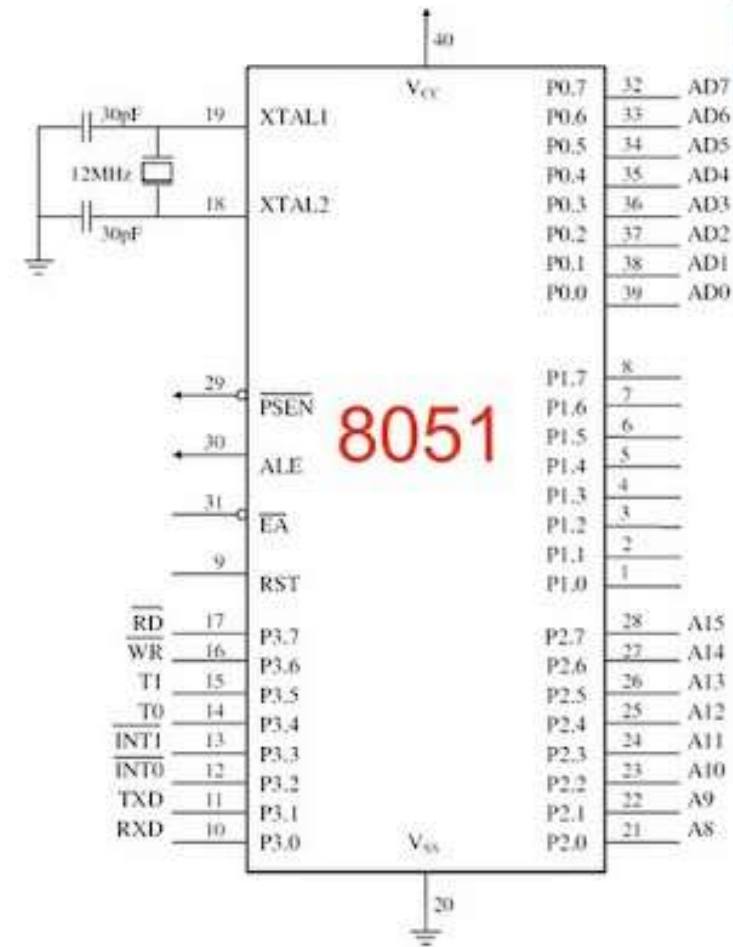
❖ Characteristics of the simple 8051:

- Does not handle floating point,
- No Cache,
- There is no MMU memory management unit,
- There is no atomic operations unit.
- There is no CPU Pipeline engineering
- Do not process commands in parallel.



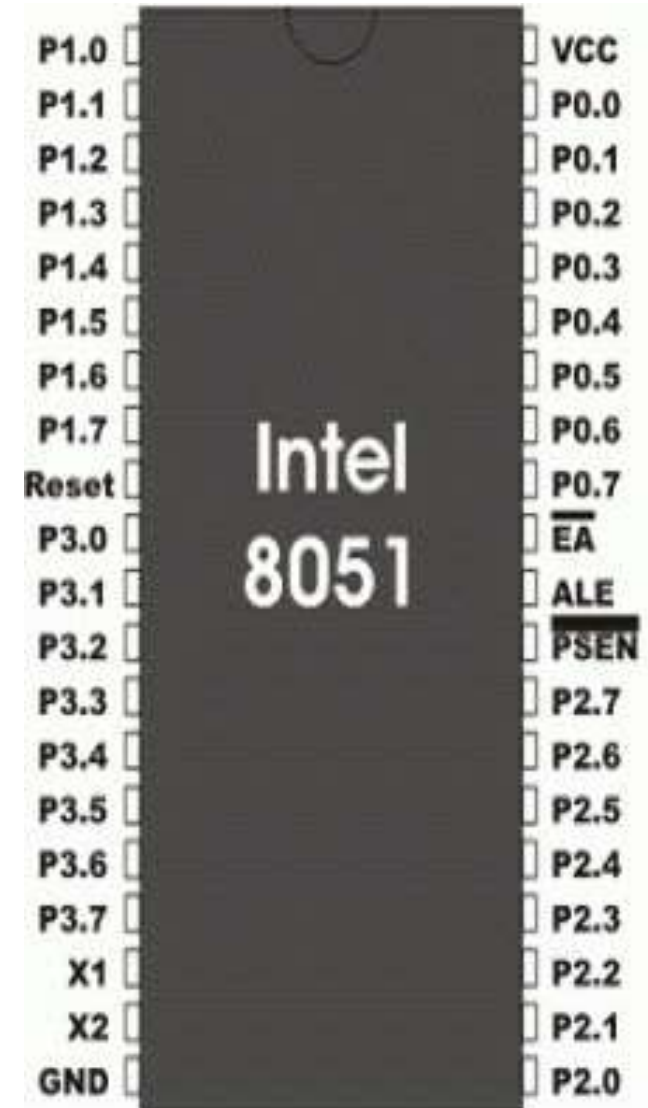
❖ Features of 8051:

- RAM size 128bytes
- The 32 bytes of RAM are used as four register sets (register-set/bank). Each bank has 8 registers.
- Stack memory/External data memory can be added up to 64 kB.



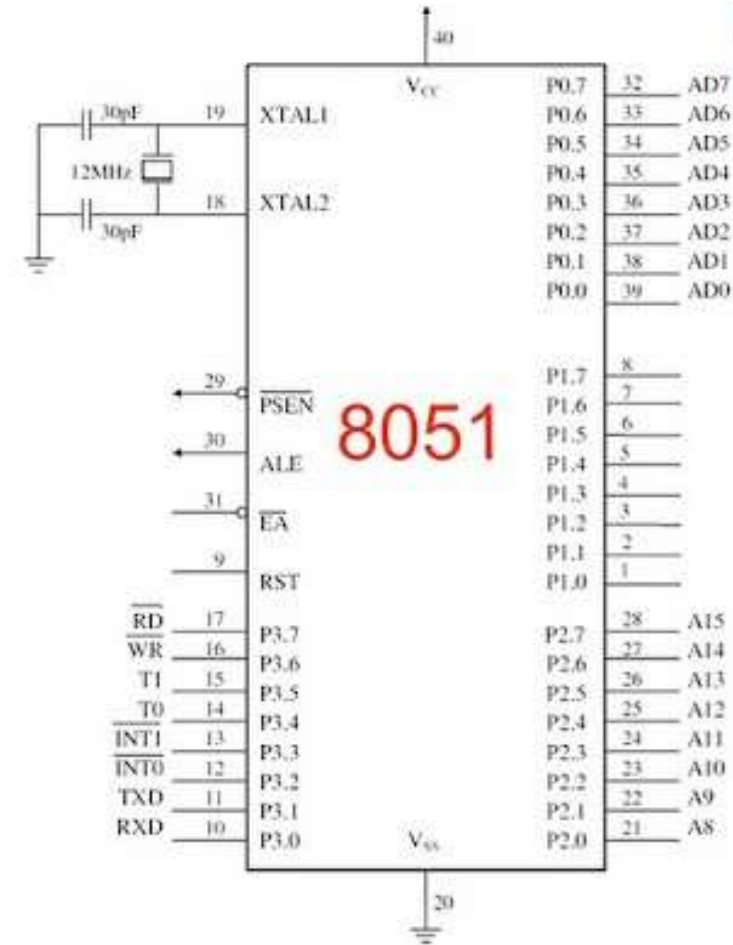
❖ Some improved versions of the 8051

- Version 8351 has on-chip ROM
- Version 8751 uses EPROM
- Version 8951 uses on-chip EEPROM or 4 kB flash memory.
- In the extended 8051 versions the address space is expanded up to 16 MB.



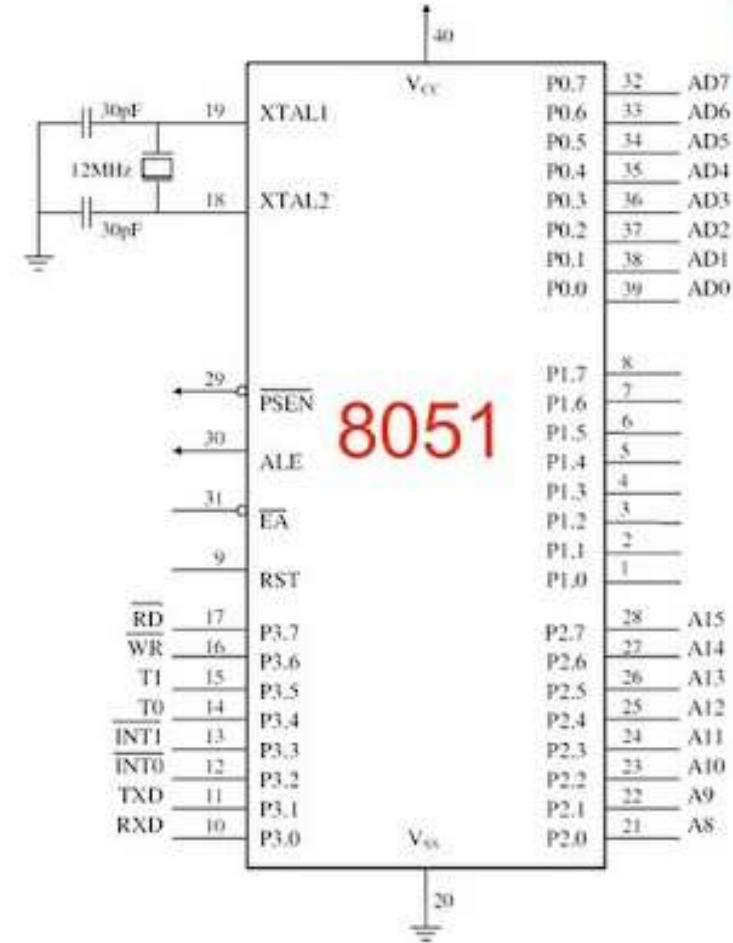
❖ Features of 8051:

- Two interrupt pins 12 and 13 INT0 and INT1.
- Four ports of 8 bits each (P0, P1, P2, P3)
- Two timers T0, T1.
- Serial Interface (SI) – programmable for three full-duplex UART modes for I/O



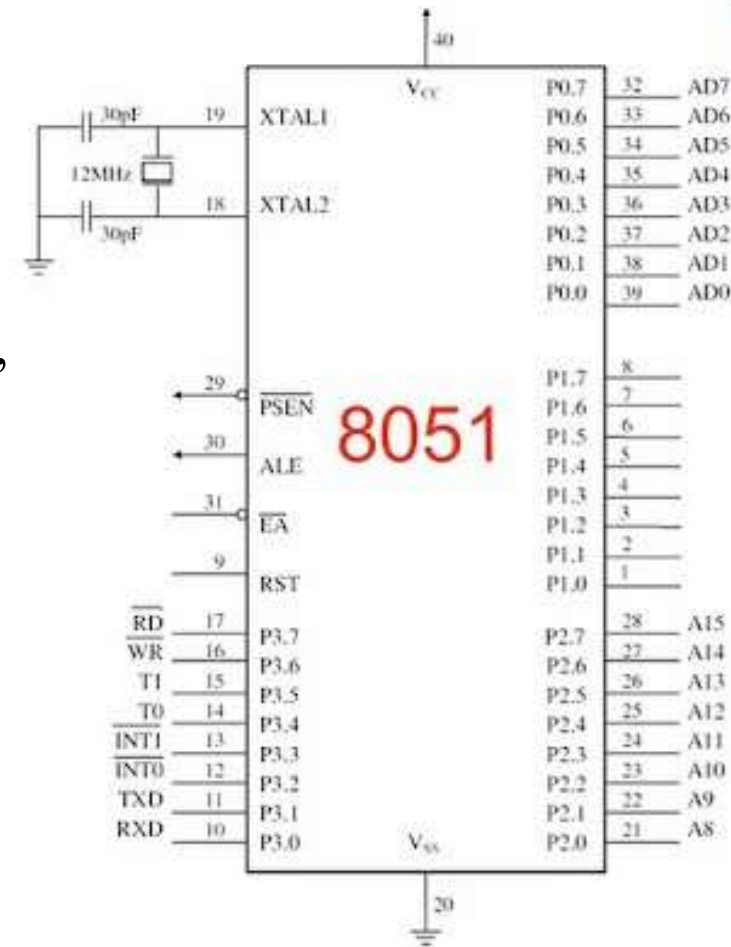
❖ Features of 8051:

- In some versions – there is a DMA controller, a pulse width modulator
- In some versions of the 8051, modems, watchdog timers, and ADCs were also integrated.
- For example, the Siemens SAB 80535-N supports an ADC with a programmable reference voltage.



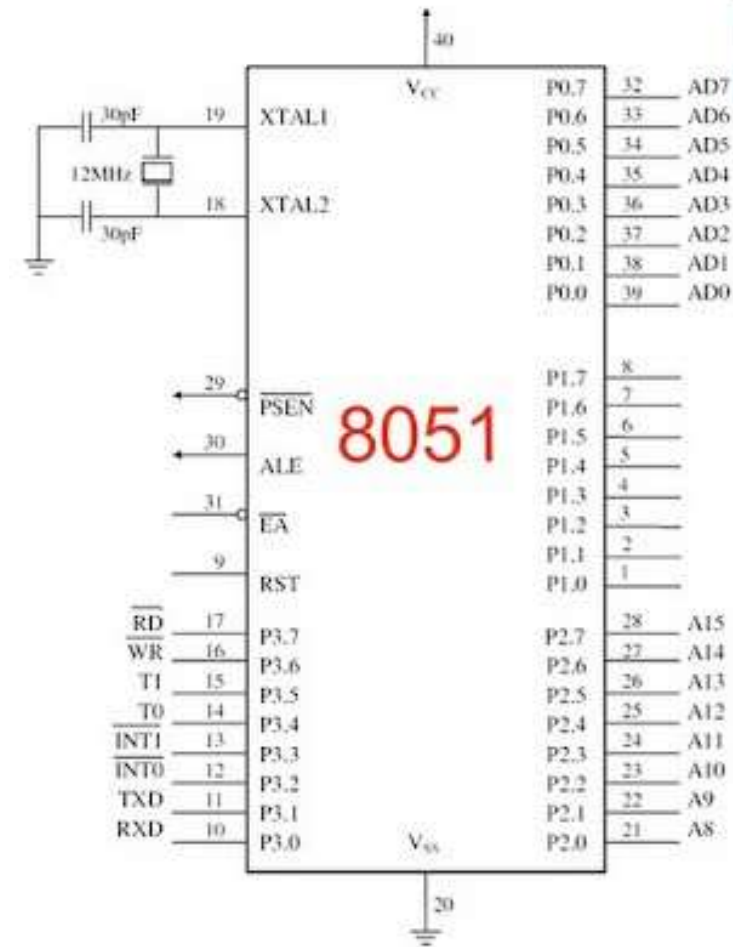
The 8051 instruction set is typically divided into the following main functional groups:

- ❖ **Data Transfer Instructions:** Performs data copying between registers, internal memory, and external memory (e.g., MOV, MOVX instructions).
- ❖ **Arithmetic Instructions:** Includes addition, subtraction, multiplication, division, increment, and decrement operations.
- ❖ **Logic Instructions:** Performs AND, OR, XOR, NOT operations and bit rotation instructions.
- ❖ **Branch Instructions:** Controls program flow such as conditional jumps, unconditional jumps, subroutine calls, and return from a cycle (e.g., LCALL, RET).
- ❖ **Boolean Processor Instructions:** This is the most prominent feature of the 8051, allowing direct manipulation of individual bits within the 210 addressable memory locations.



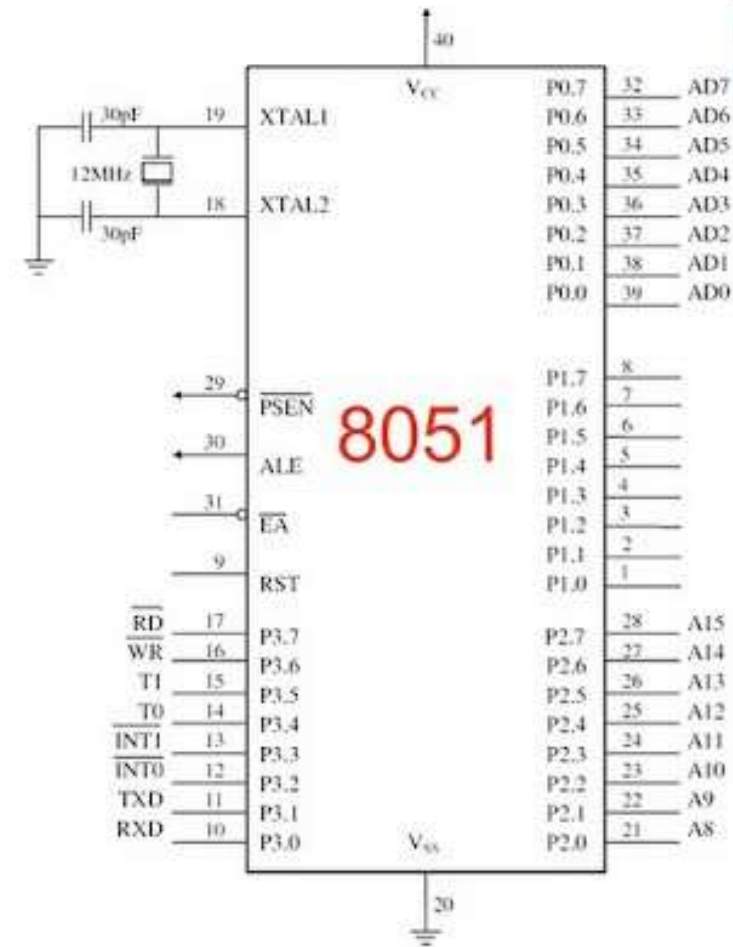
❖ Data transfer commands are:

- Move bytes between register A and another register
- Move bytes from one register/from internal RAM to another register
- Indirect move:
 - MOVC indirect
 - MOVX indirect
- Instant migration:
 - MOV immediate DPTR
- Push or Pop direct



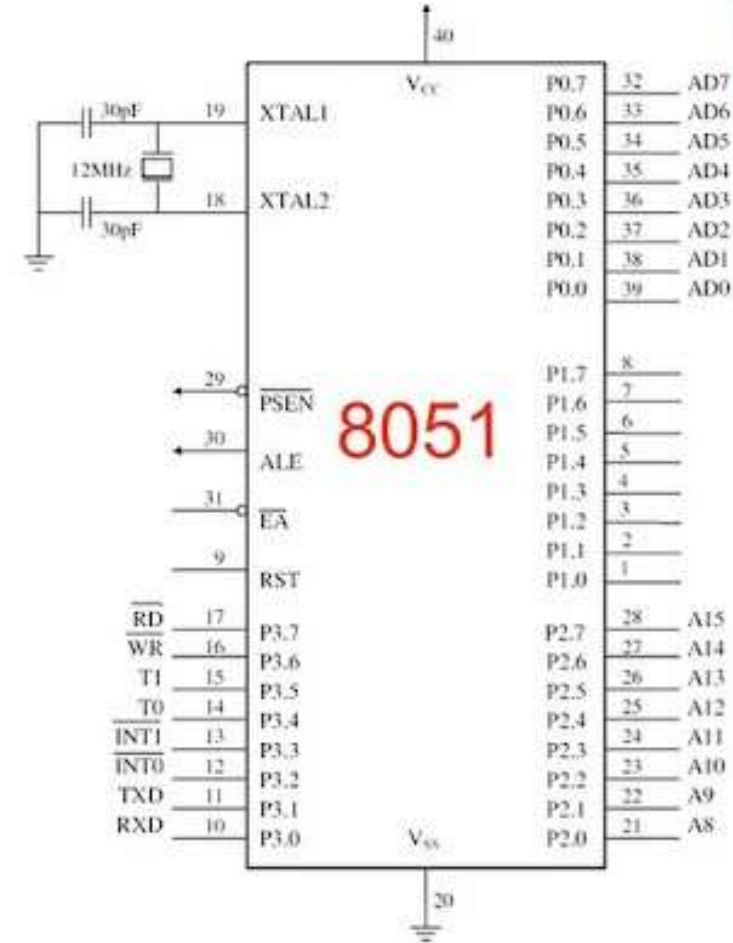
❖ Manipulates bits, bytes, and logic instructions

- Bit manipulation:
 - Set value, add value with commands like AND, OR or MOV (Clr, Set, Mov...)
- Logical command:
 - Logical instructions AND, OR, XOR
- Byte operations:
 - Delete, add or swap commands and rotation commands (CLR, RL, RC...)



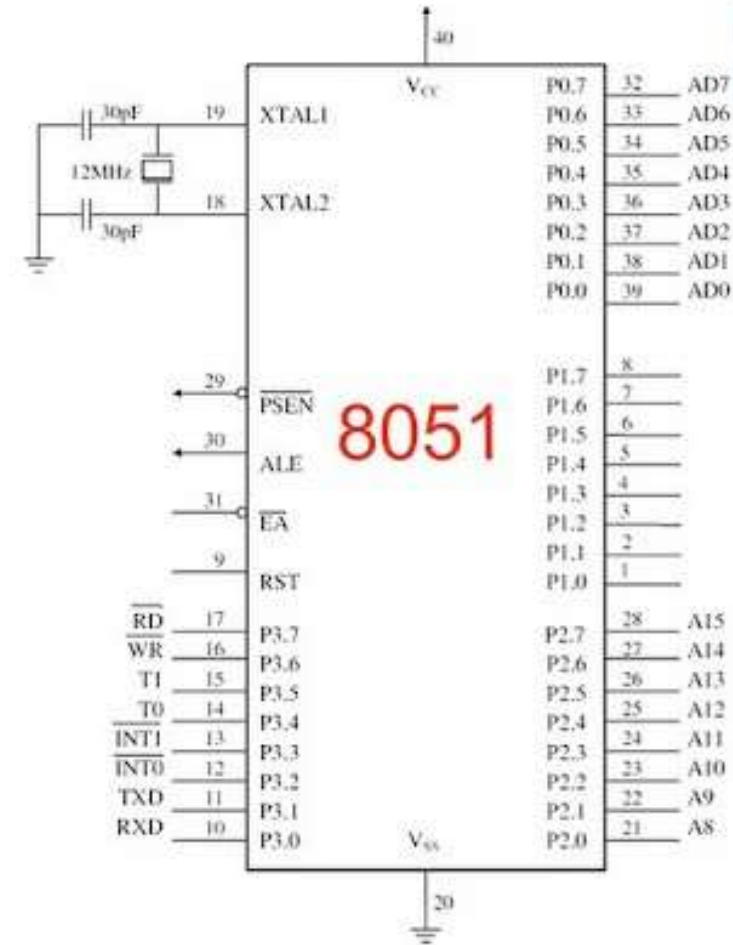
❖ Math commands:

- These are 8-bit instructions
 - Add (Add), subtract (Sub), multiply (Mul), divide (div)
- Increase or decrease orders:
 - Inc (Increment), Dec (Decrement)



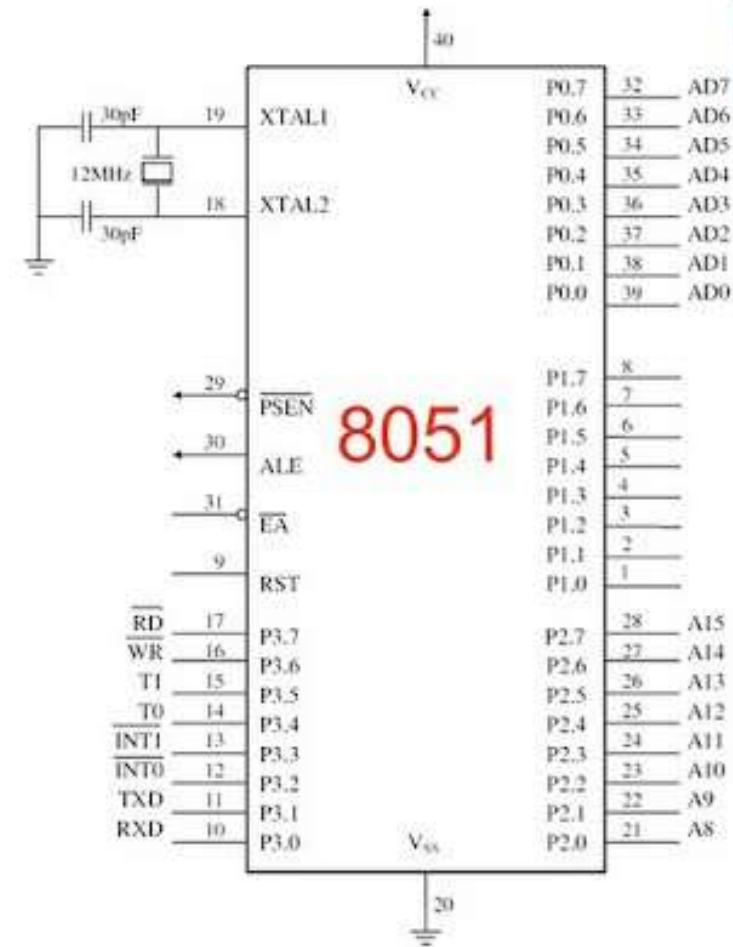
❖ Program flow control commands:

- Branching
- Conditional jump command
- Comparison command
- Subprogram call
- NOP command
- Delay command
- Break command:
 - Flow control interrupt mask bits, priority bits
 - RETI



Addressing Modes

- ❖ Immediate Addressing: The operand value is located directly within the instruction code, often used for constants defined at compile time.
- ❖ Direct Addressing: The instruction code contains the actual address of a memory location in data memory.
- ❖ Register Addressing: The operand is located in general-purpose registers (R0-R7), allowing for faster access than regular memory.
- ❖ Indirect Addressing: Uses a register (usually R0, R1, or DPTR) as a pointer to the operand's address.
- ❖ Indexed Addressing: Combines a base register and an offset value to create an effective address, very useful when accessing data arrays.



❖ 8051 has 4 input and output ports: P0, P1, P2, P3:

- It is an 8 bit input and output port
- The addresses are 0x80, 0x90, 0xA0, 0xB0 respectively
 - Each bit of port P0 is denoted P0.0 to P0.7 corresponding to addresses 0x80 to 0x87.
- Similar to P1, there are 8 ports P1.0 to P1.7 corresponding to addresses 0x90 to 0x97. P2.0 to P2.7 are addresses 0xA0 to 0xA7, P3.0 to P3.7 are addresses 0xB0 to 0xB7.
- For example:
 - MOV 0xA0, #0x0F ; means assigning value to port P2 bits is 0000 1111₂
 - INC 0xA0 ; P2 = 0000 1111₂ + 1 = 0001 0000₂

❖ P0 và P1 ports

- The pins of P0 are both the low 8 bits of the address bus and the data bus

P0

P0.0 P0.1 P0.2
P0.3 P0.4 P0.5
P0.6 P0.7
Also as
AD0- AD7

P1

P1.0 P1.1 P1.2
P1.3 P1.4 P1.5
P1.6 P1.7
Also P1.6 as
I²C clock, P1.7
as I²C serial
data, and P1.0
and P1.1 for T2
(8052)

❖ P2 và P3 ports

- The pins of P2 are the high 8 bits of the address bus

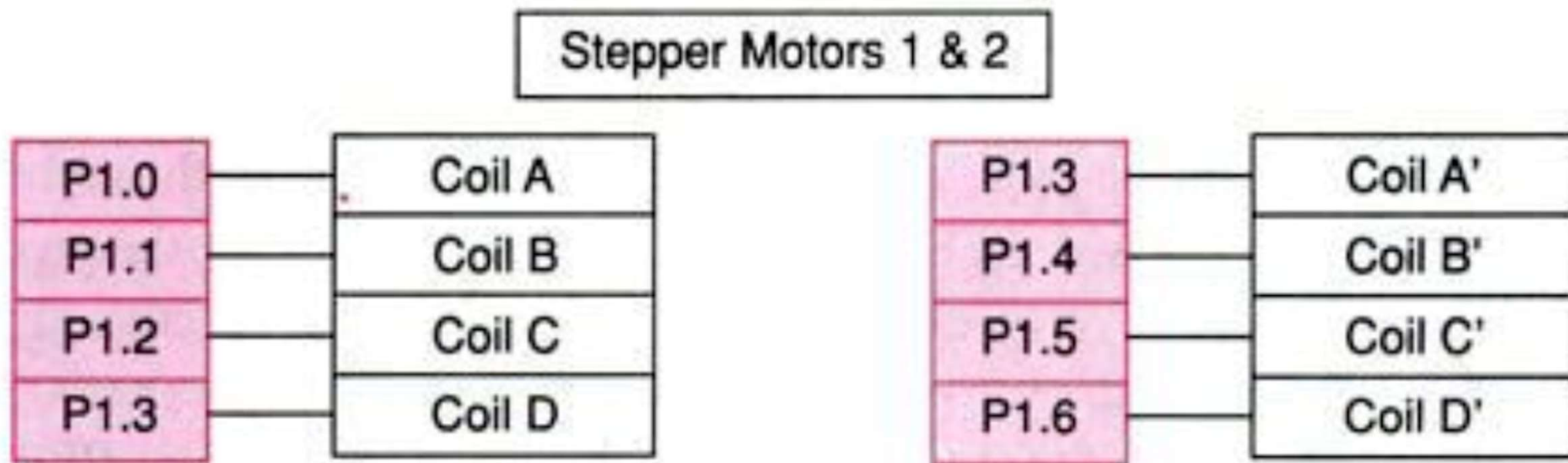
P2

P2.0 P2.1 P2.2
P2.3 P2.4 P2.5
P2.6 P2.7
Also as
A8- A15

P3

P3.0 P3.1 P3.2
P3.3 P3.4 P3.5
P3.6 P3.7
Also
RxD/SyncData,
TxD/SyncClk,
INT0/GT0,
INT1/GT1, T0,
T1, WR, RD

The circuit of input and output ports controls stepper motors used in printers



Circuit of input and output ports for six servo motors in the robot



❖ Byte programming using 8051 ports

- Byte addresses of I/O ports P0, P1, P2 and P3 of the 8051 used to access and execute read or write or other commands
- The direct 8-bit address of each address is given in the instructions
 - For example: `MOV 0xA0, #0x0F` ; means assigning values to the bits of port P2 is $0000\ 1111_2$
- The addresses of the bytes at P0, P1, P2 and P3 are 0x80, 0x90, 0xA0 and 0xB0 respectively.

❖ Byte programming using 8051 ports

- For example:
 - MOV 0xA0, #0x0F ;
 - INC 0xA0 ;

That means assigning values to the bits of port P2 is $0000\ 1111_2$

Then port P2 will be increased by 1 value

$$P2 = 0000\ 1111_2 + 1 = 0001\ 0000_2$$

❖ Bit programming using 8051 ports

- Ports P0, P1, P2 and P3 each have 8 bits
- Each bit has an address to access and perform reading or writing using bit manipulation instructions.
- Address is the bit address of the pins. Each bit address is specified in the corresponding instruction.
 - Bit addresses P0.0 to P0.7 correspond to 0x80 to 0x87.
 - Bit addresses P1.0 to P1.7 correspond to 0x90 to 0x97.
 - Bit addresses P2.0 to P2.7 correspond to 0xA0 to 0xA7.
 - Bit addresses P3.0 to P3.7 correspond to 0xB0 to 0xB7.

❖ Bit programming using 8051 ports

■ For example:

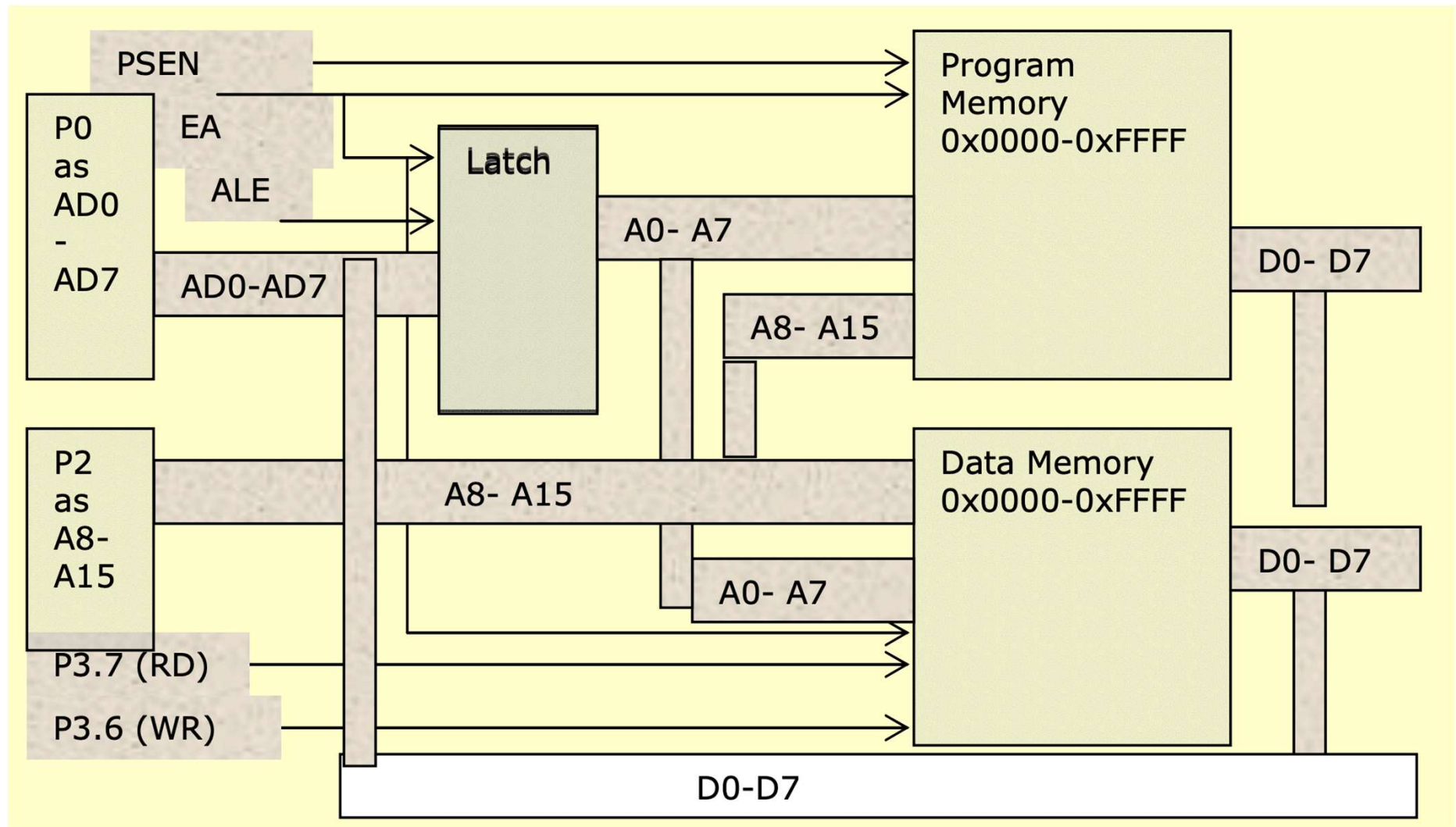
- CPL 0x90 ; Port P1 is assigned the value 0
- CLR 0x80 ; P0.0 has a value of 0.
- SETB 0x80; P0.0 has a value of 1.
- CLRB 0x80; Assign P0.0 to a value of 1.

❖ Mapping in and out of memory

- The memory and ports in the 8051 are both assigned addresses, each port has its own address range in the data memory address space.
- The communication circuitry is identically designed for the memory to connect to the external and programmable peripheral (PPI) ports.

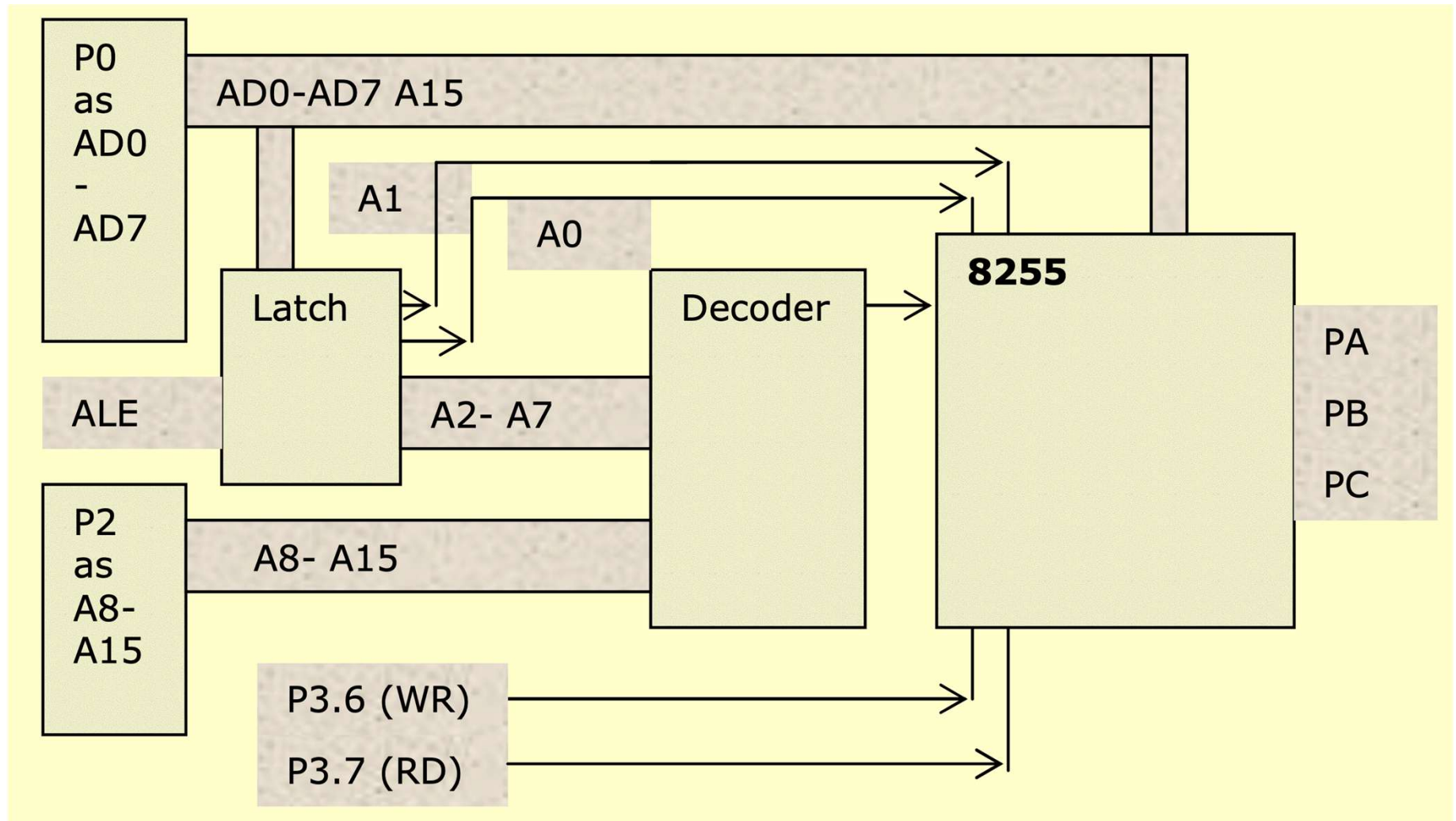
Pair with external storage

Connection to external program and data memory circuits



Pair with external storage

The interface uses the PPI external port of the 8051



❖ Port P0 in extension mode and ALE signal

- The AD0-AD7 pins of port P0 are multiplexed signals of the low address bits A0-A7 of the address bus and bits D0-D7 of the data bus.
- Time division multiplexing of signals A0-A7 and D0-D7 based on the address latch enable (ALE) processor signal

❖ Harvard memory architecture

- There are two memories — program memory and data memory.
- Two control signals — PSEN and RD to control reading from program memory or data memory.
- ALE control signal to control the use of AD0-AD7 as address or data at a given instance

❖ EA control signal

- Is a signal that when enabled — the processor always accesses external memory addresses instead of internal memory or register addresses

❖ External address RAM, Register, program memory when EA control signal is not active

- The microprocessor always accesses external memory regardless of whether the EA (Expert Advisor) is active or not
 - When the EA is connected to a low level (GND): The microprocessor will always access external program memory. In this case, internal memory (if any) will be completely ignored, and the CPU will start loading instructions from address 0000H of the external memory chip.
 - When the EA is connected to a high level (Vcc): The microprocessor will prioritize accessing internal program memory first.
- RAM and register addresses are between 0x00 and 0xFF.
- Internal program memory addresses are from 0x0000 to 0xFFFF (in the case of 4 kB internal ROM).
- External program memory addresses are 0x0000 and 0xFFFF.

❖ Counter/timer devices

- The original 8051 family had two timers T0 and T1
- The 8052 family (extension of 8051) has three timers T0, T1 and T2

❖ Counting/timing device used as a timer

- A timing device when counting inputs are provided by a clock.
- Clock pulses are given at specified intervals: acting as a timer.

❖ Counting/timing device used as counter

- A counting device when the inputs are for counting. The counter is provided with input to count from the external input pin.

- ❖ Timer T0 and T1
- ❖ TMOD special register
- ❖ TCON special register on four bits
- ❖ Four external P3 pins for external control and external counting input
- ❖ Modes 0, 1, 2, 3 of T0
- ❖ Modes 0, 1 and 2 of T1

❖ Counter/timing device External counting input in counter mode

- T0 count T0 is given input for counting from the external input T0 pin at P3.4.
- T1 counts when T1 is given input to count from external input pin T1 at P3.5.

❖ Two special registers TH1-TL1

- To access the number or time of the higher 8 bits and lower 8 bits of device T1
- Two 16-bit storage registers of the T1 device.

❖ Two special registers TH0-TL0

- To access the number or time of the higher 8 bits and lower 8 bits of device T0
- Two registers store 16 bits of device T0.

❖ **TMOD special register**

- Controls T1 and T0 modes using 4 bits per mode, programming count/timing of T1 and T0.
- One bit in each function controls whether the external port input controls or not.
- A bit that controls the function for which counter or timer mode is used.
- Two bits control the function mode of the timer/counter such as mode 0 or 1 or 2 or some other action

❖ Special register TCON: Control and status bits T1 and T0

- The four bits 4 above program the counting/timing device modes T1 and T0.
- TCON.7 and TCON.5 show the timer/counter overflow status for T1 and T0, respectively.
- TCON.6 and TCON.4 control the start and stop of the timer/counter
- The lower bits of TCON are for interrupt control for INT0 and INT1

❖ Timer/Counter T0

- 8 register bits TMOD (lower 4 bits), TCON (bits 5 and 4), TL0 (count/time bit), TH0 (count/time bit)
- Counter with input at P3.4 when bit 2 TMOD = 1, timer with internal clock timing input when bit 2 TMOD = 0
- When mode is set = 0, 8-bit Timer/Counter mode and TH0 are used and TL0 is used to pre-scale (divide) the input by 32
- When mode is set = 1, 16-bit timer/counter mode with TH0-TL0 is used for timing or counting

❖ Timer/Counter T1

- 8 bits used Register TMOD (upper 4 bits), TCON (bits 7 and 6), TL1 (count/time bit), TH1 (count/time bit)
- Counter with input at P3.5 when bit 6 TMOD = 1, timer with internal clock timing input when bit 6 TMOD = 0
- When mode is set = 0, 8-bit Timer/Counter mode and TH1 are used and TL1 is used to pre-scale (divide) the input by 32
- When mode is set = 1, timer/set mode is a 16-bit counter with TH1-TL1 used for timing or counting
- When mode is set = 2, the 8-bit Timer/Counter TH1 is used and TL1 is used to automatically reload TH1 after timeout using the preset value at TL1
- When mode is set = 3, T1 stops because TH0 is now active instead of T1.

❖ Serial Interface (SI) Function

Programming for:

- Symmetric half-duplex serial mode 0
- Asynchronous full-duplex UART mode 1, 2, or 3
- Two special 8-bit registers
 - SBUF (8-bit serial receive or transmit bit register depending on whether the instruction is using SBUF as source or destination)
 - SCON (8-bit serial mode and control bit register) and SFR PCON 7-bit

❖ SBUF

- The unique register address for the transmit and receive byte buffer when serial output or input is sent.
 - 0x99 is the address of the SI buffer.
 - The register holds the 8-bit SI line when it is written.

Example:

- | | |
|---------------|--|
| –MOV 0x99, R1 | writes to the transmit buffer data in R1 |
| –MOV R1, 0x99 | writes to register R1 data from the receive buffer |

❖ SCON

- The SI interface control register is at address 0x98.
- The three bits above program the modes as 0 or 1 or 2 or 3.
 - Mode 0 is full-duplex synchronous mode.
 - 1 or 2 or 3 is full-duplex asynchronous mode.
 - The SCON.4 bit turns the SI receiver function on or off.
 - The SCON.3 and SCON.2 bits specify which bit is transmitted and which bit is received when the mode is 2 or 3.
 - The SCON.1 bit enables or disables SI transmitter interrupts (TI) upon completion of transmission.
 - The SCON.0 bit enables or disables SI receiver interrupts (RI) upon completion of transmission.

❖ Mode 0 Function – Input or Output

- Depending on the user manual, use SBUF as source or destination
 - Synchronous serial mode data and clock input
 - Synchronous serial mode data and clock output
- Mode 0 when SCON bits 7 and 6 (mode bits) are 00

❖ Mode 1 – Input or Output

- 10 bits (start plus 8 - serial data plus stop, total 10 bits)
Serial input or output in UART mode
- Input or output whether SBUF is used for reading or writing instructions
- Baud rate can be programmed using timer T1 or T0 (T2 in 8052)
- Mode 1 when SCON bits 7 and 6 are 01

❖ Mode 2 – Input or Output

- Eleven bits (start plus 8 - serial data plus RB8 or TB8 bit plus 1 1-bit stop total) Serial input and output UART mode
- Input or output whether SBUF is used to read or write commands
- Fixed baud rate is $(f/32) \div 12$ or $(f/64) \div 12$ Mbaud/s where f = crystal frequency depending on PCON 7 bit SMOD = 1 or 0, respectively
- Mode 2 when SCON bits 7 and 6 are 10

❖ Mode 3 – SI Input or Output

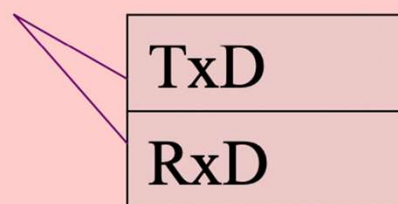
- 11 bits (start plus 8 - serial data plus RB8 or TB8 bit plus 11-bit stop sum) UART mode serial input and output
- Input or output whether SBUF is used for reading or writing instructions
- Baud rate programmable by timer T1 or T0 (T2 in 8052)
- Mode 3 when SCON bits 7 and 6 are 1,1

Serial data communication input/output

Full duplex signal, Mode 1, 2, or 3

SBUF Serial
transmit/receive data
buffer

TxD/CLK, RxD/Data pins



At receiver input from a transmitter
output

At transmitter output for a receiver
input

Processor
Processor



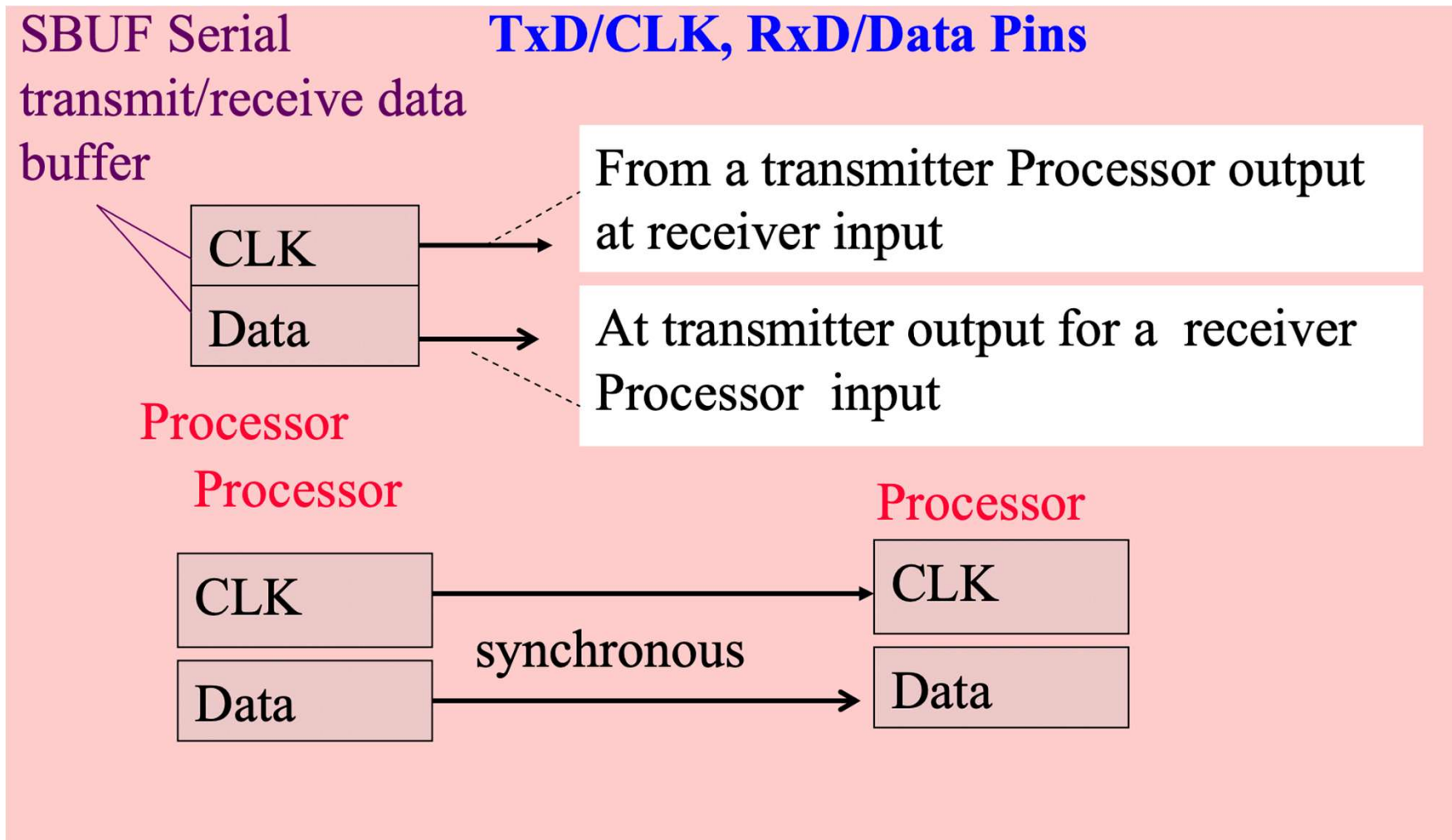
UART

Processor

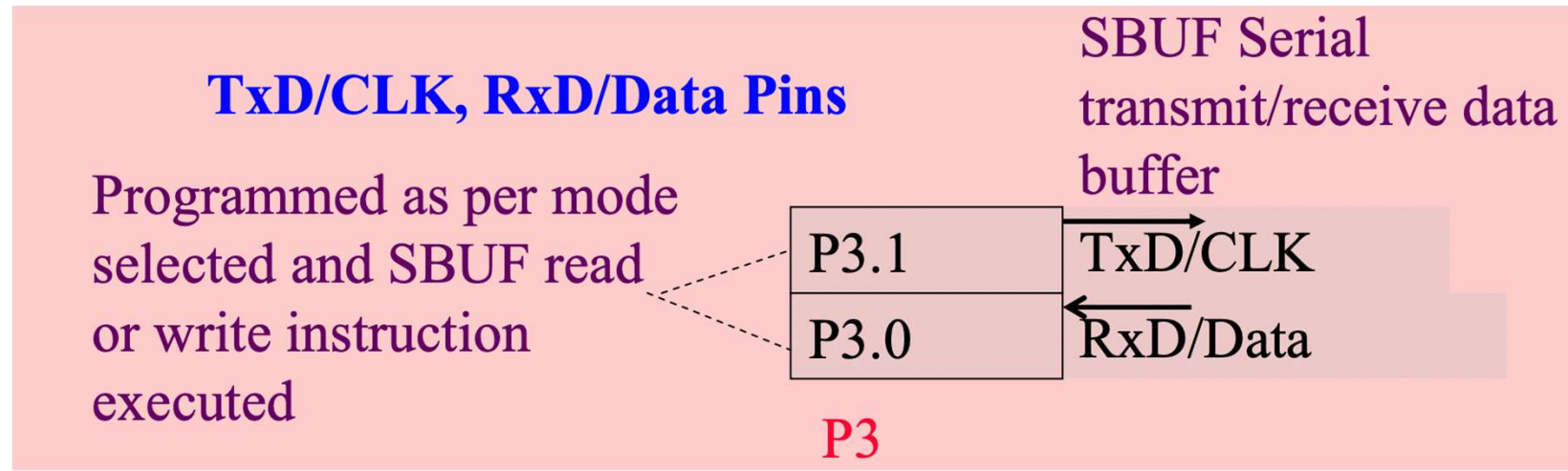


UART

Full duplex signal Mode 0



SI signals are received at the 8051 serial interface pins P3.1 and P3.0



Features of the 8051 SI

1. Mode 0 – Half-duplex synchronous operation mode
2. T8 and R8 for interprocessor communication in 11-bit format
3. Mode 1 or 2 or 3 – Full-duplex asynchronous serial communication
4. Non-programmable signals for RxD or TxD without DDR in the 8051

❖ Interrupt Sources

- External interrupts INT0, INT1
- Timer T0 and T1 overflow interrupts
- SI serial UART or synchronous mode interrupts
- SI synchronous mode interrupt (in some 8051 families)
- Timer T2 interrupt in 8052

❖ Special register SFR IE with interrupt enable bits

- Special register to mask (disable) or enable (activate) interrupts in 8051
- Programming the IE (Interrupt Enable) register at address 0xA8 for bytes
- Programming IE using bit addresses 0xA8 to 0xAF for individual bits

❖ IP Register for Priority Bits

- Interrupt priorities are set at high or low bits
- When setting a priority in IP, it overrides the default priority to execute interrupt service routines (ISRs)
 - Address 0xB8 for byte
 - Bit addresses 0x88 to 0x8C or 0x8D or 0x8E for individual bits in the register
 - An instruction can specify that a given interrupt has a high priority (=1) or low priority (=0) among the various interrupts.

❖ Features of 8051 System Interrupts

- The 8051 allows that when executing a low-priority Interrupt Service Routing Program (ISR), the program flow is interrupted when an interrupt occurs to a higher-priority Interrupt Service Routing Program (ISR).
- It allows masking all sources with a primary bit or individual sources with secondary bits by setting bits in the IE register.
- It specifies default priority levels.
- It allows overriding default priorities by setting bits in the IP register.

❖ Interrupt Vector Address

- An address from which an Interrupt Service Routing (ISR) executes up to 8 bytes or jumps to the starting address of the Interrupt Service Routing (ISR)
- When the EA bit (primary interrupt bit) is set to 1 as well as the specific interrupt bit (secondary interrupt bit) is set to 1

❖ External Hardware Interrupts INT0 and INT1

Programming for:

- Two external interrupt pins, INT0 and INT1 at P3.2 and P3.3 are used for interrupts
- P3.2 and P3.3 are used for the external interrupt pins INT0 and INT1 when bit 7 of the IE register (Enable All) bit EA is 1, and bits 0 and 2 are both 1
- Programming by TCON at the low 4 bits and bits of the IE register IE.2 and IE.0

❖ Status Bits INT0 and INT1

- TCON.3 status bit for interrupt status at INT1
- TCON.3 automatically resets to 0 when the Interrupt Service Routing Program - ISR starts servicing interrupt INT1.
- TCON.1 status bit for interrupt status at INT0 and TCON.1 automatically resets to 0 when the Interrupt Service Routing Program - ISR starts servicing interrupt INT0.

❖ The INT0 and INT1 edges or control bits for level types:

- TCON.2 for the interrupt type at INT1 and has a value of 1 if it is a non-zero edge-triggered type.
- TCON.0 for the interrupt type at INT0 and it has a value of 1 if it is a non-zero edge-triggered type.

Hard interrupt of the improved 8051 version

Vector addresses and default priority

INT0	0x0003
T0	0x000B
INT1	0x0013
T1	0x001B
Serial	0x0023
T2	0x002B
Syn Serial	0x0053 in few versions

High

Hardware priority default (The priority setting will be assigned to the software in the IP register and override the default value)

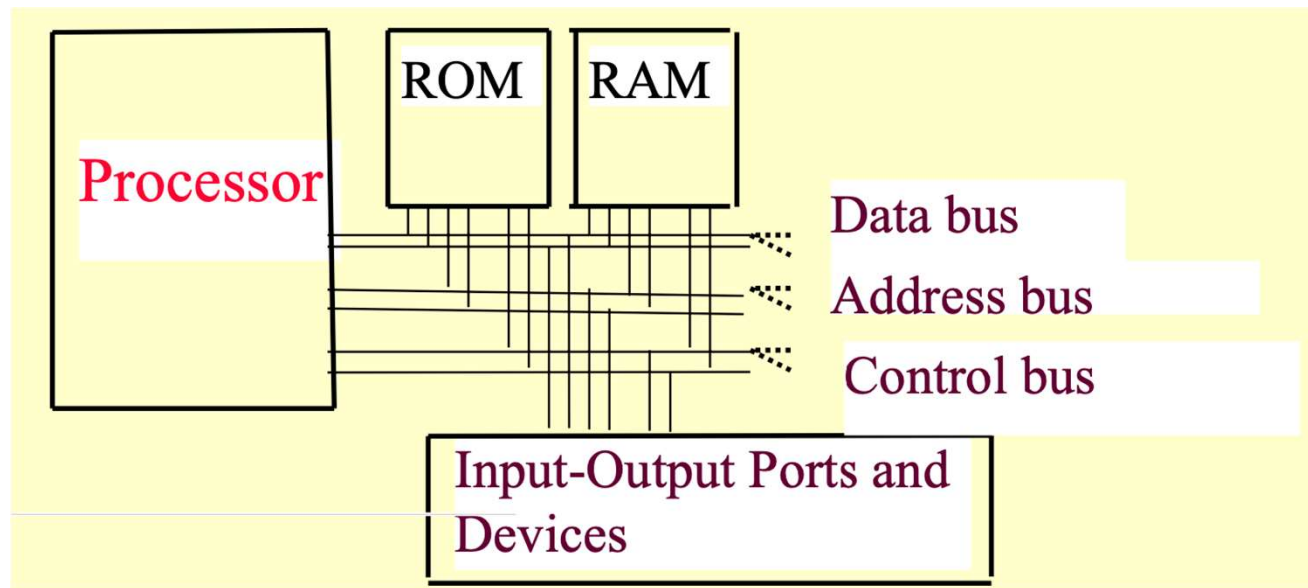
Low

❖ Communication using the System Bus

- The interface between the processor, memory, and I/O devices uses the memory system bus.
- System bus – is a system of connections for a simple bus architecture with three sets of signals.
- System bus – defined by the address bus, data bus, and control bus (A, D, C).
- The design of the system bus interface follows the timing scheme of the processor signals, the speed, and the length of memory words for data and instructions.

❖ Communication using the System Bus

- Internal and external buses of the microprocessor
 - Distinguishing features in the system



Communication for a simple bus architecture

Communication of the processor, memory, and I/O devices using the memory system bus

❖ Communication using the System Bus

■ Address Bus

- The processor provides the address of each byte or word of the instruction to the memory system via the address bus.
- The processor's execution unit, when requested, provides the address of the data (byte or word) to read or write to memory via the address bus.
- The 32-bit address bus is used to retrieve instructions or data from an address specified by the 32-bit number.

❖ Communication using the System Bus

■ Address Bus : Example

- To start the processor, it resets the program counter PC to address 0.
- The processor places address 0 on the address bus, and the instruction at address 0 is retrieved from memory when address 0 is reset.
- To retrieve the instruction, it needs to load register r1 from memory address M.
- The processor assigns address M on the address bus, and the data at address M is loaded.

❖ Communication using the System Bus

■ Address Bus : Example

- The processor assigns address m to an instruction; it fetches the instruction via the data bus from address m . [For a 32-bit instruction, the word at the data bus is from addresses m , $m+1$, $m+2$, and $m+3$.]
- The instruction executes to save register bits $r1$ to memory address M ; the processor assigns address M on the bus and sends the data at address M via the data bus. [For 32-bit data, the word at the data bus is sent to memory addresses M , $M+1$, $M+2$, and $M+3$.]

❖ Communication using the System Bus

■ Data Bus

- Instruction Fetch — The processor is given the address of the instruction, and it receives the instruction back via the data bus.
- Data Read — When given the address of the data, it loads the data via the data bus.
- Data Write — When given the address of the data, it stores the data in memory via the data bus. A 32-bit data bus fetches, loads, or stores 32-bit instructions or data.

❖ Communication using the System Bus

■ Control Bus

- Generates timing control signals for operations during communication.
- Synchronizes all subsystems.
- Activates Address Strobe (ALE) [Address Strobe (AS) or Valid Address (ADV)]
- Read (RD) or write (WR) memory or read (IOR) or write (IOWR) peripheral I/O or valid data (DAV)
- Other control signals depending on the processor.

❖ Communication using the System Bus

- DMA and Interrupt Control Signals
 - Interrupt Acknowledgment (INTA) [on a request to the processor from an event]
 - INT (Interrupt) from an external device to the system
 - Hold Acknowledgment (HLDA) [from an external request to enable the use of system buses]
 - HOLD when an external device sends a hold request to access direct memory (DMA).

❖ Communication using the System Bus

■ DMA and Interrupt Control Signals

Example:

- The processor issues the address, it also issues a control signal to read memory and waits for data or instructions.
- Memory stores instructions or data during the time the memory read signal is active (unless the processor disables it).
- The processor assigns the address on the address bus and places data on the data bus, then issues a control signal to write memory (after allowing sufficient time for all data bits to set up) to save the signal to memory.
- Memory must write (store) data during the time the memory write signal is active (unless the processor disables it).

❖ Communication using the System Bus

- Program memory and data bus access are multiplexed in the Harvard Architecture
 - Address and data buses are multiplexed.
 - The PSEN control signal operates when accessing program memory using the address and data buses.
 - The READ or WRITE control signal operates when accessing data memory using the address and data buses.

❖ Communication using the System Bus

- Time Division Multiplexed (TDM) Data and Address Bits
 - TDM — Time division multiplexed: Different time intervals have different sets (channels) of signals.
 - Address signals in one time interval t
 - Data bus signals in another time interval.
 - The interface circuit for demultiplexing the bus channels uses a control signal in such systems.
 - Control signal: Address Strobe (ALE) in 8051, Address Strobe (AS) in 68HC11, and Address Validity (ADV) in 80196.
 - ALE, AS, or ADV are used to demultiplex the address and data buses to the devices.

❖ Communication using the System Bus

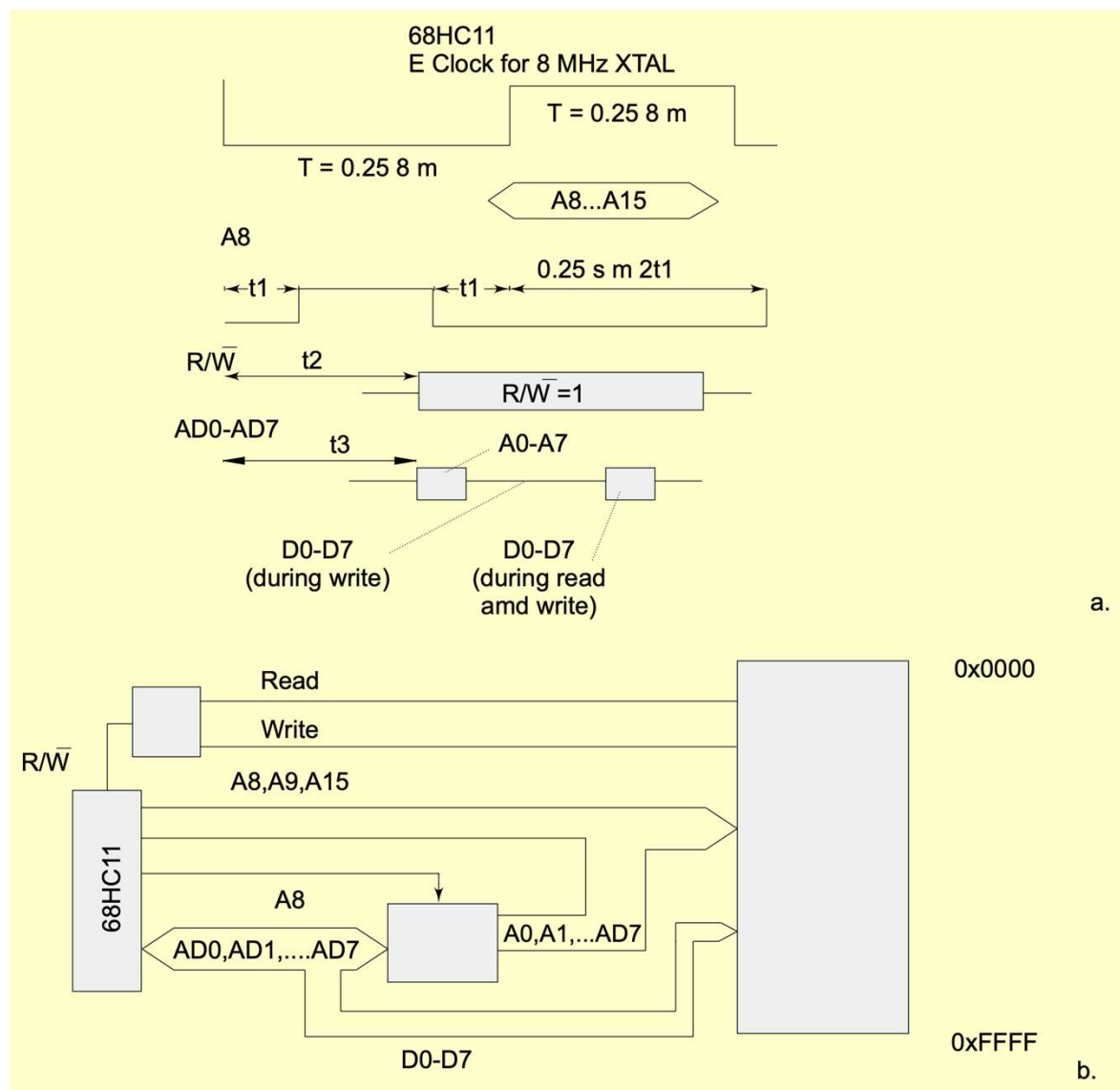
- The interface circuit uses Latch and a decoder
 - ALE to latch the address.
 - PSEN to read program memory using the data bus address.
 - Each chip of the memory or processor interface has a separate chip select input from the decoder.
 - The decoder is a circuit that takes the appropriate address bus signals at the input and the control circuit signals to generate the corresponding CS (chip select) control signals for each device (memory and port).

❖ Communication using the System Bus

■ *Mạch giao tiếp*

- Bao gồm bộ chốt Latch, bộ giải mã và bộ tách kênh
- Được thiết kế theo tín hiệu điều khiển có sẵn và lược đồ thời gian của tín hiệu bus.
- Mạch kết nối tất cả các thiết bị, bộ xử lý, bộ nhớ và thiết bị IO thông qua các bus hệ thống.
- Còn được gọi là mạch keo - *glue circuit* sử dụng khi kết nối các thiết bị và bộ nhớ với bus hệ thống và bộ xử lý
- Có thể được thiết kế bằng GAL (generic array logic) hoặc FPGA

❖ 68HC11 bus signal and memory communication

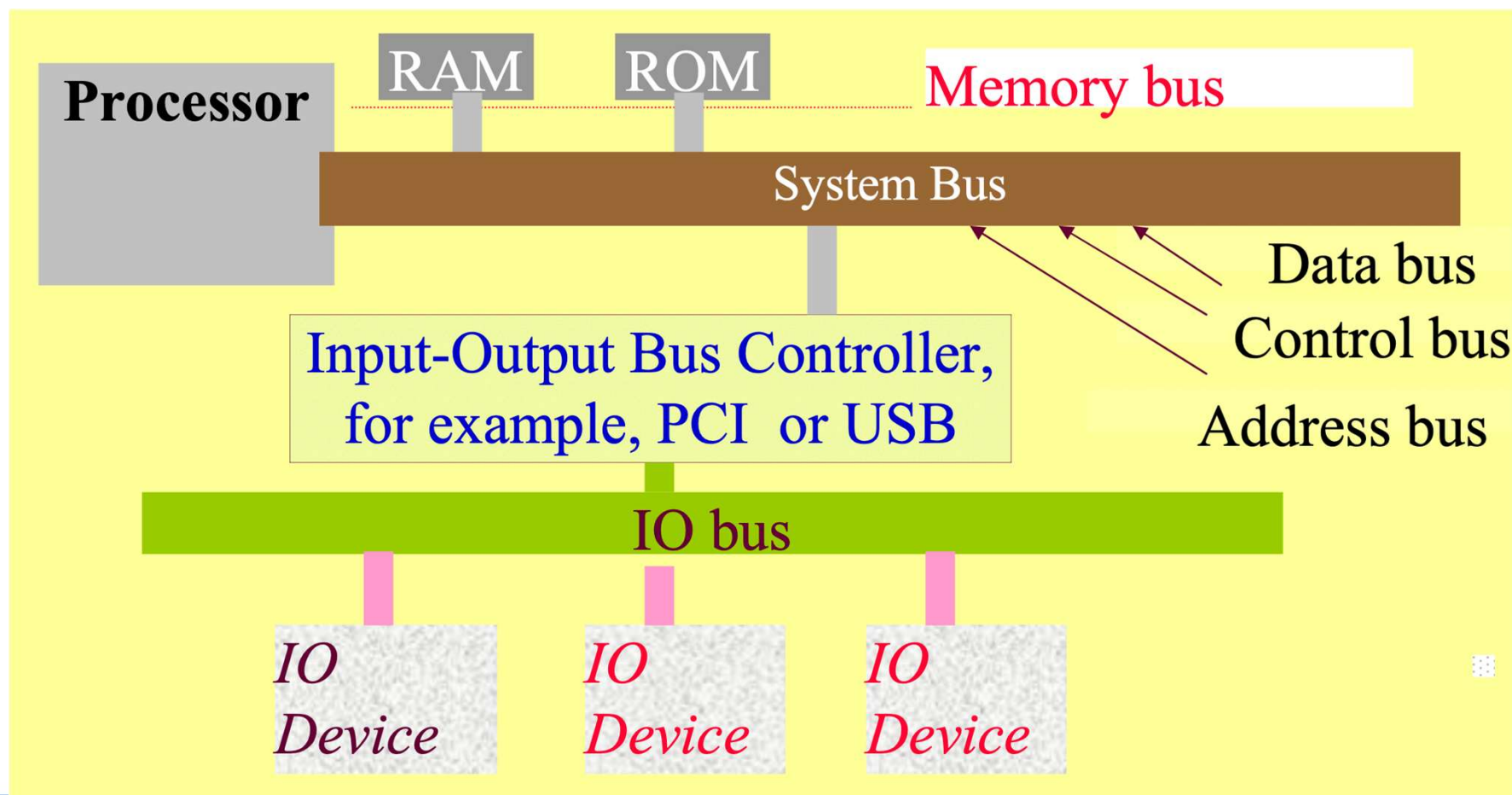


❖ Communication uses the system bus and I/O

- System bus connections:
 - Processor
 - Memory system and subsystems
 - Another set of signals called the I/O bus
 - The processor's interface with the system bus at the first level and the I/O bus at the second level

❖ Communication uses the system bus and I/O

- Memory system bus and I/O bus connections in a bus architecture



❖ Multilevel Bus

- Separate memory and I/O buses using PCI and USB bus controllers; separate disk I/O bus.

